

Evaluating AI Detection Models for Social Media Content



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Evaluating AI Detection Models for Social Media Content

A Study of Text-Based Tools, Challenges, and Opportunities in
Moderating AI-Generated Content

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Management Summary

This thesis systematically evaluated the effectiveness of widely used AI detection tools (Grammarly, GPTZero, and ZeroGPT) in identifying AI-generated content on social media platforms like X (formerly Twitter), where posts are brief and linguistically informal. The evaluation combined a benchmarking study using 1,600 tweets generated by various AI models (GPT-2, GPT-3.5, GPT-4o) and human-authored content with in-depth qualitative insights from expert interviews.

Key findings revealed that, contrary to initial assumptions, more sophisticated generative AI models such as GPT-4o were easier for current detection tools to identify compared to earlier models like GPT-2. While older models produce less coherent text, they also exhibit erratic or unnatural linguistic patterns that overlap with low-quality human writing, complicating classification. In contrast, advanced models exhibit distinctive and consistent patterns, which detection methodologies, particularly transformer-based and hybrid approaches, can exploit effectively (Zellers et al., 2019; Solaiman et al., 2019). GPTZero and ZeroGPT, which utilize advanced, multi-faceted detection frameworks, consistently outperformed Grammarly, which leans more on statistical heuristics (Elkhatat et al., 2023).

Qualitative insights from expert interviews provided critical context, revealing widespread skepticism about the long-term viability of AI detectors. Experts emphasized that the detection landscape is constrained by the "arms race" dynamic: each advancement in generative AI reduces the reliability of existing detection tools. Ethical concerns such as the risk of false positives in academic or journalistic contexts and privacy limitations related to data access and behavioral tracking were identified as key barriers (Glikson & Woolley, 2020; Elkhatat et al., 2023). Furthermore, there is increasing concern over the feedback loop where detectors train on data increasingly influenced by generative AI, further complicating accuracy.

Based on these findings, several practical recommendations were made. These include the adoption of hybrid and modular detection frameworks that integrate linguistic, syntactic, and deep learning techniques (Fagni et al., 2021; Kumarage et al., 2023), especially for use on platforms where short-form content dominates. Tools should incorporate metadata and behavioral signals, such as posting patterns and device data to enhance contextual understanding and reliability (Rossi et al., 2022; Yang et al., 2024). Ensuring transparency through explainable AI (Rai, 2019), retraining models regularly to counter concept drift (Akram, 2021; Weber-Wulff et al., 2023), and promoting collaboration among academic, industrial, and regulatory stakeholders were also highlighted as critical.

In conclusion, while current AI detection tools demonstrate measurable effectiveness against certain models, their long-term reliability is uncertain due to the relentless progression of generative AI. Future-proof moderation strategies must remain adaptive, ethically grounded, and multi-dimensional recognizing that detection success today does not ensure success tomorrow.

Preface

This study is the outcome of my master's thesis in the master's program Information Management at Tilburg University. This master's thesis report focuses on evaluating the effectiveness of AI detection models in identifying AI-generated content on social media platforms, particularly within the context of short-form text on X (formerly Twitter). This thesis marks the conclusion of an important chapter in my academic journey and signifies the final step in my time as a student. I deeply appreciated my years at Tilburg University, and I am excited to take the next step forward into the professional world.

Writing this thesis was not always easy, but I remained determined to complete it. The process offered valuable insights, and I have learned a great deal during this time. I believe the findings and knowledge in this study can be of use for future developments in the field of AI content moderation. I would like to thank my supervisor from Tilburg University, Dr. Shiva Shekhar, who provided me with support, constructive feedback, and important guidance throughout the writing of this thesis. I also appreciate the time and dedication it took to bring this research to completion. Without the help of Dr. Shiva Shekhar, it would have been much more challenging to finish this thesis.

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1. Introduction

1.1. Problem Indication

The rise of large-scale generative artificial intelligence (AI) models is reshaping how digital text content is produced and consumed on social media. Language models (e.g., GPT-4) can produce text that is increasingly difficult to distinguish from human-created content. This phenomenon introduces opportunities for automated content creation and personalized marketing but also fosters challenges concerning authenticity, credibility, and the potential for malicious uses such as misinformation, spam, or identity impersonation.

Within the Information Management discipline, detecting AI-generated text has emerged as a critical issue. Traditional digital forensics and data quality principles must extend to newer methods that identify linguistic or structural markers unique to machine-generated text. Scholars highlight the importance of explainable AI (XAI) in ensuring stakeholders understand and trust AI-driven decisions (Rai, 2020). Meanwhile, platforms that leverage network effects the added value derived from massive user data must adapt or risk allowing deceptive content to flourish (Gregory et al., 2021).

Beyond these technical considerations, trust in AI-driven detection is paramount (Glikson & Woolley, 2020). Users may distrust such systems if legitimate posts are wrongly removed (false positives) or harmful content goes undetected (false negatives). Recent studies also find that some machine-learning (ML) models, originally trained to assess text quality, can inadvertently rate AI-generated text more favorably than human-authored content (Bevilacqua et al., 2024). These issues underscore the need for robust, transparent, and scalable detection solutions to accurately label AI-generated text on social media.

Although academic interest in generative AI detection is growing, current text-detection models still struggle with obfuscation (e.g., paraphrasing) and rapidly evolving generative techniques. Given time constraints, this study will concentrate solely on text-based detection, forgoing an exploration of image/video methods. Focusing on text alone allows a deeper examination of linguistic and structural challenges inherent to moderating AI-generated content, thereby filling a key gap in systematically evaluating the strengths and limitations of leading text-detection models.

Given these emerging challenges, X (formerly Twitter) becomes an especially relevant platform for this study due to its short-form, rapid-fire communication style. Tweets are limited to around 280 characters, leading to concise messages that contain fewer linguistic markers for detection models to analyze. The frequent use of hashtags, emojis, and informal abbreviations can further mask or distort patterns linked to AI-generated text. In addition, the platform's real-time nature amplifies content quickly; trending topics can proliferate in minutes, heightening the potential harm of AI-driven misinformation and demanding swift, accurate detection methods. Many influential figures, politicians, journalists, and prominent personalities actively use X, raising the stakes whenever deceptive content circulates. Research has also shown that the inherent brevity of tweets can undermine purely semantic-based detection approaches, underscoring the need for stylistic or structural markers. Stylometric signals such as punctuation patterns, phraseology, and linguistic diversity can be critical in detecting AI-authored content, especially when adversaries employ obfuscation tactics like slang, paraphrasing, or sudden shifts in writing style (Kumarage et al., 2023).

Overall, by assessing the performance of various text-detection algorithms, this study aims to offer actionable insights to academia, platform moderators, and policy makers. It is especially relevant to the fields of Digital Forensics, Data Governance, and Systems Strategy, given the prevalence of AI-generated content on social media. Through a focused evaluation on X, the research seeks to illuminate how these next-generation technologies influence organizational practices, user trust, and public discourse and how detection tools must evolve to meet the growing threat of AI-driven manipulation.

1.2. Problem Statement

Sophisticated generative AI tools can now produce content that is nearly indistinguishable from human-generated text, creating a critical detection gap on social media platforms. This gap is especially evident on X where short-form, rapid-fire posts make AI-generated text both easy to distribute and difficult to detect. Existing detection solutions often struggle to remain accurate across diverse content types and can degrade under adversarial attacks or shifts in domain context (Ara et al., 2024). This detection gap affects multiple stakeholder groups:

- **Platform stakeholders** such as social media moderators, policy enforcers, and community managers require accurate tools to flag questionable posts.
- **Users and communities** face risks from misinformation and deepfake content, which can undermine trust and authentic engagement particularly on platforms like X where information spreads instantly.
- **Researchers and developers** need deeper insights into detection limitations while exploring advanced algorithms.
- **Regulatory bodies** must address AI-driven disinformation campaigns with scalable, transparent solutions.

A comprehensive evaluation of text-based AI detection tools is critical to mitigating these risks to contextualize how textual analyses fit into larger content moderation strategies. Such an approach includes:

- **Empirical Benchmarking:** Measuring accuracy, precision, recall, and other relevant metrics on diverse, labeled datasets of AI- and human-generated text.
- **Analysis of Linguistic Markers:** Determining how detection models respond to paraphrasing, slang, short-form posts, or minimal editing.
- **Recommendations for Improvement:** Proposing enhanced feature integration (e.g., metadata or user behavior signals) and robust design principles (e.g., transparency, interpretability).

This study will focus on publicly available or widely used AI detection models, primarily targeting short-form text on social media. However, certain limitations must be acknowledged:

1. **Rapid Evolution:** Generative models advance swiftly, meaning any detection findings may quickly become outdated.
2. **Data Access:** Ethical considerations and privacy regulations can constrain dataset size or authenticity.
3. **Tool Opacity:** Proprietary detection algorithms often do not disclose underlying methods, challenging in-depth assessments.

By thoroughly examining the strengths and limitations of existing detection models, this thesis aims to provide practical recommendations for researchers and platform stakeholders, ultimately contributing to a more secure and reliable digital ecosystem.

1.3. Research Question

The main question of this research is:

“How effective are selected AI detection models in accurately identifying AI-generated generated posts on X, and how can these models be enhanced in relation to linguistic and structural markers to address current limitations and evolving generative techniques?”

1.3.1. Sub-Questions

To address the main research question, the study will explore the following sub-questions:

Detection Model Performance:

- Which AI detection tools are most relevant and widely used for short-form text on social media platforms?
- How do these AI detection models perform in identifying AI-generated text on X?

Linguistic and Structural Markers:

- Which linguistic and structural markers (e.g., punctuation, phrase length, stylometric signals) are most indicative of AI-generated text on X?

Challenges and Limitations:

- What are the primary challenges detection models face in identifying AI-generated text, particularly in adversarial settings or evolving generative techniques?

Enhancement Strategies:

- What potential improvements can be made to existing detection models to enhance their effectiveness in identifying AI-generated text?

1.4. Research Design

This study adopts an exploratory-evaluative research design, combining quantitative benchmarking of selected AI detection models with qualitative insights from industry experts. By focusing on X (formerly Twitter) and its short-form text environment, the design ensures a targeted and context-specific examination of detection effectiveness.

Quantitative Component (Benchmarking):

1. Model Selection and Setup

- Tool Identification: Begin by selecting a small set of widely used or publicly available AI detection tools (e.g., GPTZero, Copyleaks, Originality.AI). Selection criteria will include relevance to short-form text, availability of an API or dashboard, and popularity among users and researchers.
- Platform Focus: Emphasize X as the primary data source, ensuring the tools are tested on short-form, potentially slang-heavy content.

2. Dataset Compilation

- Collect AI-Generated Samples
- Gather Human-Authored Samples.

3. Data Analysis

- Perform statistical comparisons (e.g., ANOVA, paired t-tests) to identify significant differences in tool performance.
- Identify common failure modes (e.g., slang usage, partial paraphrasing) by analyzing incorrect predictions.

Qualitative Component (Expert Interviews):

This study adopts also a qualitative approach, semi-structured interviews to capture expert perspectives on AI text-detection. Experts will be selected from academic researchers with experience in AI detection. Interviews will focus on challenges, trust issues, and recommendations for improving AI-generated content detection.

1.5. Research Method

The study follows a two-phase methodology, combining experimental testing with qualitative insights:

Phase 1: Experimental Testing

1. Dataset Collection:
 - a. AI-Generated Samples: Generate or collect short-form posts from state-of-the-art language models (e.g., GPT-4) fine-tuned to produce text in Twitter's 280-character style.
 - b. Human-Authored Samples: Curate genuine tweets from public repositories or X's API, ensuring a diverse range of topics, user demographics, and writing styles.
 - c. Data Validation: Manually inspect a subset of both AI- and human-authored texts to verify label accuracy and ensure authenticity.
2. Tool Evaluation:
 - a. AI detection tools will be assessed by running them against the collected dataset and recording performance metrics.
 - b. Statistical analysis (e.g., confusion matrices, ROC curves) will be applied to compare tool effectiveness across various test conditions.

Phase 2: Qualitative Insights

Interviews

In-depth discussions will be conducted with AI researchers who possess direct experience in natural language processing, machine learning, or content moderation. These semi-structured interviews will examine current detection challenges, model limitations, and opportunities for enhancement, particularly within short-form text environments like X (formerly Twitter). Specific topics include integration hurdles (e.g., limited transparency in proprietary tools), linguistic nuances (slang, abbreviations), and future directions (novel linguistic markers, ensemble methods, interpretability). By encouraging participants to provide detailed, firsthand insights, this format captures the practical realities of developing and deploying AI text-detection solutions in dynamic social media contexts.

Ethical Considerations

The study will adhere to ethical guidelines related to data privacy and informed consent. Any sensitive content in datasets will be anonymized to prevent ethical breaches.

2. Literature overview

2.1. Overview of generative AI applications on social media

Generative Artificial Intelligence (AI) has significantly transformed social media platforms, influencing content creation, user engagement, and marketing strategies. This section provides an overview of generative AI applications in social media.

Generative AI techniques are fundamentally altering how content is produced and curated on social media platforms. AI-powered tools can generate text, images, and videos, enabling users to create unique and engaging content with ease. For instance, Adobe Firefly utilizes generative AI to assist users in crafting visually appealing media, thereby enhancing the quality and diversity of content shared online. This advancement democratizes content creation, allowing individuals without extensive technical skills to produce professional-grade materials (Kumar & Yadav, 2024).

The integration of generative AI into social media has led to more personalized user experiences. AI algorithms analyze user behavior and preferences to generate tailored content recommendations, fostering increased engagement and satisfaction (Bashiri & Kowsari, 2024).

Generative AI also impacts communication dynamics on social media. AI-driven chatbots and virtual assistants facilitate interactions, sometimes emulating human conversational styles. However, this raises concerns about the authenticity of online interactions and the potential for misinformation. Research indicates that AI-generated content can influence users' opinions and behaviors, underscoring the need for transparency and ethical considerations in deploying these technologies (Jakesch et al., 2023).

Despite its benefits, the application of generative AI in social media presents challenges, particularly concerning misinformation and ethical implications. The ease of generating realistic yet fabricated content can lead to the spread of false information, necessitating robust detection and regulation mechanisms. Researchers emphasize the importance of developing standards and policies to address these issues, ensuring the responsible integration of AI technologies into social media platforms (Pan et al., 2024).

2.2. AI-Based Text Detection Approaches

Machine Learning-Based Approaches

Machine learning (ML) techniques have been widely employed to detect AI-generated text by analyzing linguistic structures and statistical irregularities. These approaches rely on pre-defined handcrafted features that can capture the subtle differences between AI-generated and human-authored text. Traditional machine learning models, such as decision trees, support vector machines (SVMs), and logistic regression, analyze various textual attributes, including perplexity scores, sentence structure anomalies, coherence metrics, and word frequency distributions, to differentiate between AI and human text. Despite their effectiveness in specific cases, these models often struggle with advanced AI-generated text that closely mimics human writing. Baraheem and Nguyen (2023) applied a Convolutional Neural Network (CNN)-based classifier to detect AI-generated text by identifying linguistic inconsistencies in textual datasets. Their model achieved high accuracy by recognizing patterns such as unnatural sentence structuring and repeated phrasing commonly seen in AI-generated text. However, one of the key challenges highlighted in their study is the requirement for large labeled datasets, which are often scarce.

Similarly, Shu et al. (2020) proposed an adversarial deep learning framework that analyzes contextual relationships within text, leveraging transformer-based models such as BERT and RoBERTa. Their findings suggest that these models can effectively differentiate AI-generated text by detecting irregular coherence patterns and unnatural grammatical formations. However, as AI-generated text becomes

more fluent and indistinguishable from human writing, the effectiveness of such models diminishes. Both studies underscore the limitations of machine learning-based approaches, particularly in detecting highly sophisticated AI-generated content that mimics human fluency and writing styles. While ML models provide interpretable and computationally efficient solutions, their reliance on manually engineered features makes them less adaptable to emerging AI text-generation techniques. Moreover, AI-generated content on X often exhibits characteristics that blend seamlessly with human-written posts, further reducing the effectiveness of traditional machine learning-based detection methods.

Deep Learning-Based Approaches

Deep learning, particularly Transformer-based architecture, has emerged as a powerful tool for AI-generated text detection due to its ability to capture intricate linguistic patterns and contextual dependencies. Unlike traditional machine learning approaches that rely on handcrafted features, deep learning models automatically learn patterns from large-scale datasets, making them more adaptable to evolving AI-generated content. These models analyze text at a granular level, detecting statistical anomalies, contextual inconsistencies, and linguistic artifacts that distinguish AI-generated text from human-authored content. The use of self-attention mechanisms in Transformer models enables them to identify long-range dependencies within text, which is crucial in differentiating AI-generated text from human writing, particularly in social media contexts where content is concise and highly contextual. Zellers et al. (2019) developed Grover, a model specifically designed to both generate and detect fake news. Their research demonstrated that large-scale transformer models outperform humans in detecting AI-generated content by analyzing probability distributions and linguistic variations. They found that AI-generated text tends to have higher consistency in sentence structure and phrase repetition, making it easier for transformers to detect than for human readers. However, a key limitation of this approach is its dependence on access to model-generated probability distributions, which may not always be available when dealing with proprietary AI models. Additionally, Grover itself is an AI text-generation model, raising ethical concerns regarding its potential misuse in generating misleading or deceptive content.

Solaiman et al. (2019) explored the fine-tuning of transformer-based detection models using adversarial training to enhance their robustness. Their findings revealed that while AI-generated text often mimics human writing well, it frequently overuses high-probability words and lacks the variation seen in human writing. The study demonstrated that BERT-based classifiers exhibit strong contextual understanding, making them highly effective in detecting AI-generated content. Despite their effectiveness, deep learning models face several practical challenges in AI-text detection. One major drawback is the computational expense associated with training and deploying these models. Transformer-based architecture requires vast amounts of labeled data and significant processing power, making real-time detection on social media platforms like X challenging. Additionally, the continuous evolution of AI-generated text necessitates frequent retraining of detection models to maintain their efficacy. This presents logistical challenges, particularly for platforms with high content turnover, where AI-generated misinformation can rapidly spread before detection models are updated.

Both studies by Zellers et al. (2019) and Solaiman et al. (2019) highlight the strengths of deep learning-based approaches in AI-text detection. They emphasize the need for models that not only analyze linguistic patterns but also adapt to the dynamic nature of AI-generated content. While deep learning methods provide a more sophisticated and accurate detection framework compared to traditional machine learning models, their reliance on extensive datasets, high computational requirements, and access to AI-generated text probability distributions limit their scalability. In the context of X and other social media platforms, these challenges become even more pronounced, as real-time detection requires lightweight, efficient models capable of processing vast amounts of data while maintaining high accuracy.

Feature-Based Approaches

Instead of relying solely on deep learning, some researchers have focused on linguistic feature engineering, where stylistic and semantic markers are extracted to distinguish human and AI-generated text. These approaches utilize syntactic analysis, burstiness measures, and word frequency distributions to detect inconsistencies. Jawahar et al. (2020) examined the statistical traces left by transformers in generated text, revealing that contextual word embeddings can serve as reliable indicators of AI-generated content. Their study demonstrated that transformer-generated text follows a more uniform sentence dependency structure than human-written text, making it possible to develop feature-based classifiers for detection. However, they also noted that highly trained AI models reduce these statistical traces, making detection more challenging over time.

Ippolito et al. (2020) further investigated linguistic patterns in AI-generated text, identifying key statistical markers such as overuse of common words, repetitive phrasing, and a lack of stylistic diversity. Their research found that AI-generated text tends to exhibit flatter probability distributions in language modeling, making it distinguishable from human-written content. This feature-based approach offers a scalable and computationally efficient detection method. However, the study also found that its effectiveness diminishes when dealing with shorter text samples, where distinguishing features are less prominent. These studies suggest that while feature-based approaches provide an alternative to deep learning models, they require continuous adaptation as AI-generated text becomes more human-like in structure and coherence.

Hybrid Approaches

Given the limitations of both deep learning and feature-based methods, recent research has explored hybrid approaches that integrate multiple detection techniques to enhance performance. Fagni et al. (2021) conducted a large-scale study on AI-generated misinformation, demonstrating that machine-generated fake news exhibits distinct syntactic and semantic structures compared to human-written misinformation. Their research proposed a hybrid detection method that combines traditional machine learning, deep learning, and feature-based techniques, achieving higher accuracy than individual methods. By leveraging both statistical markers and transformer-based models, their approach successfully identified AI-generated text across different datasets. However, hybrid methods introduce additional complexity, requiring more computational resources and extensive feature engineering.

2.3. Review of AI-Generated Text Detection Tools

While various detection approaches such as machine learning, deep learning, and linguistic analysis form the foundation of AI-text detection, their practical implementation in detection tools determines their real-world effectiveness. These tools, often built on multiple approaches, are used to identify AI-generated content in different contexts, including social media platforms like X.

Akram (2021) conducted a large-scale evaluation of multiple AI-text detection models assessing their effectiveness across different types of AI-generated content. The study found that no single detection method is entirely reliable, as AI models continuously evolve to produce more human-like text. A key challenge identified was concept drift, where detection models degrade in performance over time due to advancements in AI text generation. The study emphasized the necessity of continuously updating detection methodologies to counteract these adaptations and maintain detection accuracy, particularly in dynamic online environments like social media.

Cui et al. (2023) conducted a study on AI-text detection in social media environments, focusing on the challenges posed by short-form content. Their research revealed that existing detection models struggle to identify AI-generated tweets and comments due to their brevity, lack of contextual depth, and human-like phrasing. The study underscores the limitations of current AI detection tools in adapting to the fast-paced, informal, and highly dynamic nature of social media language. Their findings highlight the

urgent need for detection models that can effectively process platform-specific linguistic patterns and short-form textual structures, particularly in real-time settings like X.

Elkhataat et al. (2023) conducted a comprehensive evaluation of commercial AI content detection tools, assessing their accuracy, bias, and reliability. Their findings revealed significant shortcomings, particularly the tendency of these tools to falsely classify human-written text as AI-generated, leading to concerns about false positives in academic and journalistic contexts. Additionally, the study highlighted a lack of transparency in detection methodologies, as many commercial tools operate as black boxes without disclosing their inner workings. These limitations raise critical concerns about the dependability of AI detectors. In social media environments like X, where misinformation spreads rapidly, unreliable AI detection tools could inadvertently misidentify real human-generated content, leading to misinformation suppression or censorship issues. The study calls for greater transparency and standardization in AI detection tools, particularly for platforms that rely on automated moderation systems.

Kumarage et al. (2023) investigated the effectiveness of stylometric analysis in detecting AI-generated text by examining writing style variations. Their study found that AI-generated content often exhibits uniform sentence structures and limited lexical diversity, making it distinguishable through stylometric techniques. They demonstrated that AI text lacks the natural variability seen in human writing, which can be leveraged for detection. However, their findings also highlight a key limitation: stylometric methods perform better on long-form text and struggle with shorter passages, such as tweets, where writing patterns are less pronounced, posing challenges for AI detection on platforms like X.

Lastly, Weber-Wulff et al. (2023) evaluated AI-generated text detection tools, highlighting their limitations in identifying sophisticated AI-generated content. Their study revealed that detection tools struggle with newer models, especially when text is paraphrased or modified, a critical issue for X's dynamic content. They emphasized the need for continuous retraining and real-time updates to maintain detection accuracy. The study also suggested a multi-faceted approach, integrating various detection methods for improved reliability. In the context of social media, their findings stress the urgent need for adaptive detection frameworks to counter AI-generated misinformation effectively.

AI-generated text detection remains a complex and evolving challenge. Detection models struggle with short-form content, where limited context makes differentiation difficult. Issues such as false positives, bias, and the lack of transparency in commercial tools further complicate the detection process. Additionally, static detection methods quickly become outdated as AI models improve, necessitating continuous adaptation. To enhance reliability, detection frameworks must integrate multiple approaches and undergo frequent retraining. As AI-generated content continues to advance, the development of more robust and adaptive detection solutions will be essential in maintaining content authenticity online.

2.4. Challenges in detecting AI-generated content

The rapid advancement of generative artificial intelligence (AI) has significantly impacted social media platforms, enabling the creation of content that closely resembles human-generated material. This development poses substantial challenges for detection mechanisms tasked with distinguishing between human and AI-generated content. Kondamudi et al. (2023) provide an extensive analysis of fake news phenomena on social networks, emphasizing the complexities involved in identifying misleading information due to its multifaceted characteristics. Their study outlines foundational theories, diverse perspectives, and key features that typically characterize fake news, including sensationalism, lack of credible sources, emotional manipulation, and deliberate misinformation strategies designed to provoke public reaction and amplify content virality. These attributes often mirror the stylistic and linguistic features found in AI-generated misinformation, thereby complicating its detection. Furthermore, Kondamudi et al. highlight the rapid and viral dissemination of misinformation on platforms such as Twitter (X), where real-time interactions and short-form content intensify these detection challenges.

Consequently, distinguishing between authentic human-generated content, AI-generated misinformation, and traditional fake news becomes increasingly complex, requiring more sophisticated, adaptive, and context-aware detection methodologies.

Modern AI systems, such as OpenAI's GPT series and Meta's LLaMA, have achieved remarkable proficiency in generating text and images that are virtually indistinguishable from those created by humans. This sophistication complicates the detection process, as traditional markers of inauthenticity become less apparent. A study by Rossi et al. (2022) investigated the ability of individuals to distinguish between real and AI-generated social media profiles. In an experiment involving 375 participants, the researchers found that fully AI-generated profiles, including posts written by advanced text generators, were difficult for humans to identify, supporting the hypothesis that detecting such content is extremely challenging for the average user (Rossi et al., 2022).

Existing detection tools often struggle to keep pace with the rapid evolution of AI-generated content. Research by Weber-Wulff et al. (2023) evaluated multiple AI-text detection models on various AI-generated content types and analyzed detection performance over time. The study revealed that no single detection method is entirely reliable, as AI models continuously evolve to produce more human-like text. A key challenge identified was concept drift, where detection models degrade in performance over time due to advancements in AI text generation, emphasizing the necessity for continuous retraining to maintain detection accuracy.

The widespread availability of generative AI tools has led to an influx of AI-generated content across various platforms. A study by Yang et al. (2024) conducted a systematic analysis of X accounts using human faces generated by Generative Adversarial Networks (GANs) for their profile pictures. The researchers identified 1,420 such accounts engaged in activities like spreading scams, spam, and amplifying coordinated messages. They estimated that between 0.021% and 0.044% of active Twitter users, approximately 10,000 daily active accounts utilize GAN-generated profile images, underscoring the emerging threats posed by multimodal generative AI.

AI-generated content raises significant ethical and privacy issues, including the potential for identity theft, security breaches, and the spread of misinformation. The ease with which AI can produce realistic, yet fabricated content necessitates robust detection and regulation mechanisms to mitigate these risks. A study by Elkhataf et al. (2023) evaluated AI-generated text detection tools, highlighting their limitations in identifying sophisticated AI-generated content. The study revealed that detection tools struggle with newer models, especially when text is paraphrased or modified, a critical issue for dynamic content on social media platforms. The researchers emphasized the need for continuous retraining and real-time updates to maintain detection accuracy and suggested a multi-faceted approach integrating various detection methods for improved reliability. Different social media platforms face unique challenges in detecting and managing AI-generated content. For example, in the lead-up to the U.S. presidential election, platforms like Facebook were criticized for failing to consistently label AI-generated images of political candidates, highlighting the difficulties in implementing effective detection and labeling systems. This inconsistency underscores the need for platforms to develop standardized approaches to identify and disclose AI-generated content to maintain user trust and ensure the integrity of information (Six, 2024).

2.5. Knowledge Gap

Despite significant advancements in generative AI and extensive research efforts aimed at detecting AI-generated content, the literature review highlights several critical knowledge gaps that remain unaddressed, particularly in the context of short-form content on social media platforms like X.

Firstly, although existing studies have extensively examined various AI-generated text detection methodologies, ranging from traditional machine learning techniques (Baraheem & Nguyen, 2023; Shu et al., 2020), transformer-based deep learning (Solaiman et al., 2019; Zellers et al., 2019), to feature-

based and hybrid approaches (Fagni et al., 2021). There remains limited comprehensive evaluation of these methods specifically in short-form and informal contexts typical of social media platforms. Tweets and similar short messages pose unique detection challenges due to their brevity, informal language usage, frequent use of emojis, slang, abbreviations, and hashtags, which significantly diminish the effectiveness of traditional linguistic and syntactic markers employed by most detection methodologies (Cui et al., 2023; Kumarage et al., 2023). Thus, a systematic comparison of detection tool performance specifically tailored to short-form textual content remains relatively under-explored.

Secondly, although the literature frequently acknowledges the importance of integrating non-textual data such as metadata, behavioral analytics, and cross-platform user interactions (Rossi et al., 2022; Yang et al., 2024), there is still a notable scarcity of empirical studies evaluating the practical implications and effectiveness of such integrative methods in real-world settings. Most current AI detection methodologies rely predominantly on textual analyses alone, without sufficiently exploring how incorporating metadata or user behavioral signals might enhance reliability, particularly in detecting coordinated disinformation campaigns or automated bot-driven content.

Thirdly, there is a clear gap concerning the adaptability of AI detection models to the rapidly evolving nature of generative AI systems. Studies consistently identify the issue of "concept drift," where detection model effectiveness degrades over time due to improvements and adaptations in generative models (Akram, 2021; Weber-Wulff et al., 2023). Yet, few empirical studies systematically address how frequent detection models should be retrained or updated, what retraining strategies are most effective, or how robust these retraining methods are against adversarial or obfuscation techniques increasingly employed by advanced generative models.

Furthermore, despite considerable emphasis on the ethical implications and risks of misclassification, such as false positives, which could inadvertently censor authentic human-generated content. There remains insufficient exploration of strategies to mitigate these risks, particularly within sensitive domains such as journalism, academia, or legal contexts (Elkhatat et al., 2023; Rai, 2019). Research on the practical integration of explainable AI (XAI) techniques into AI-generated text detection tools is limited, leaving stakeholders without clear guidance on implementing transparent and accountable detection practices.

Finally, cross-sector collaboration between academia, industry, and regulatory bodies has been repeatedly suggested as essential for developing effective AI detection standards and policies (Pan et al., 2024; Six, 2024). However, there is still a notable absence of detailed research into how such collaborative frameworks can be practically structured, managed, and regulated, particularly given the competing interests of privacy, innovation, and ethical governance in AI applications.

3. Theoretical framework

3.1. Conceptual model

Based on the findings from the literature review, a conceptual model has been developed to illustrate the relationships among critical factors affecting the effectiveness of AI-generated text detection on X (formerly Twitter). This conceptual model provides a theoretical foundation explaining how AI-generated content indirectly influences the effectiveness of AI-text detection through the chosen detection methodology. This framework is visually represented in Figure 1.

The model begins with the independent variable, AI-generated content on X. Contemporary generative AI systems such as GPT and LLaMA produce short-form text content characterized by high fluency, coherence, and stylistic sophistication, closely resembling human-authored messages. This realistic and human-like quality significantly complicates the task of distinguishing AI-generated from authentic human-generated tweets. As AI-generated content evolves, becoming more nuanced and linguistically diverse, detection methods must advance in parallel, adopting increasingly sophisticated approaches. Thus, the nature of AI-generated text inherently motivates advancements in detection methodologies.

The core mediating factor is the detection methodology, representing the technical approaches and analytical techniques employed to identify AI-generated content. Detection methodologies broadly encompass machine learning, deep learning (particularly transformer-based models), and hybrid frameworks that blend statistical measures with deep neural networks. Each methodology presents unique analytical capabilities, strengths, and limitations that directly influence detection performance. For instance, deep learning-based methods, especially transformer architectures, exhibit superior performance in identifying complex linguistic and contextual patterns indicative of AI-generated text, outperforming simpler statistical or rule-based models (Zellers et al., 2019; Shu et al., 2020). Thus, the continual advancement and refinement of detection methodologies are crucial in effectively addressing the evolving sophistication of AI-generated content.

Eventually, the model identifies the effectiveness of AI-generated text detection as the dependent variable. This effectiveness is defined through key metrics such as accuracy, precision, recall, and error rates (false positives and false negatives). The effectiveness of detection tools is significantly influenced by their underlying detection methodology, as more advanced methodologies are better equipped to accurately differentiate AI-generated tweets from human-written ones. Given the complexity and evolving nature of AI-generated content, tools employing sophisticated analytical techniques, such as hybrid methods combining transformer-based linguistic analysis with statistical indicators are expected to demonstrate higher detection accuracy.

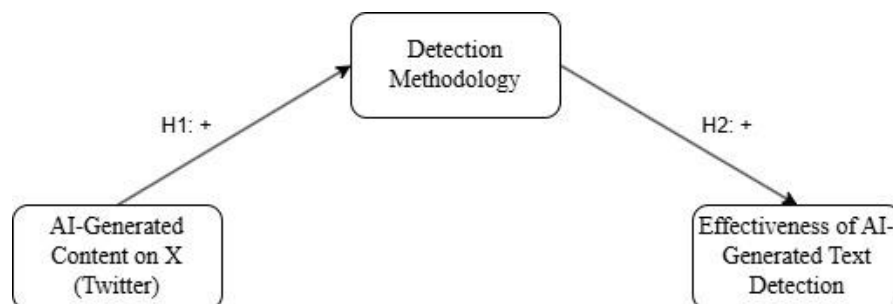


Figure 1: Illustration of the conceptual model

3.2. Hypothesis development

Based on the established conceptual framework, two central hypotheses guide the empirical investigation of this research. The relationships articulated in these hypotheses align with the conceptual model illustrated in Figure 1, emphasizing the mediating role of detection methodology.

The first hypothesis addresses the relationship between the sophistication of AI-generated content and the selection of detection methodologies. This hypothesis posits that as AI-generated content grows more sophisticated, becoming increasingly fluent, coherent, and human-like the need arises for more advanced detection methodologies (Shu et al., 2020; Zellers et al., 2019). In other words, simpler detection methods might suffice for content generated by earlier or less advanced models, such as GPT-2, but the emergence of highly advanced generative models like GPT-4.5 necessitates more complex detection frameworks, such as deep learning architectures or hybrid methodologies. Consequently, it is hypothesized that increased content sophistication positively influences the choice and requirement of more advanced detection methods:

H1: AI-generated content on X positively influences the advancement of detection methodologies.

The second hypothesis examines how the choice of detection methodology impacts the overall effectiveness of AI detection tools. Drawing from the existing literature (Shu et al., 2020; Zellers et al., 2019), it is anticipated that methodologies characterized by greater complexity, such as hybrid approaches combining linguistic, statistical, and neural features, or advanced deep learning architectures like Transformers provide superior performance. Specifically, these sophisticated methodologies should achieve higher accuracy and precision while reducing false positives and false negatives compared to simpler machine learning-based or statistical approaches. Thus, the second hypothesis is stated as follows:

H2: The sophistication of the detection methodology positively influences the effectiveness of AI-generated text detection tools.

By testing these hypotheses using empirically derived datasets and rigorous benchmarking procedures, this study aims to provide a robust evaluation of how detection methodologies mediate the relationship between AI-generated content and detection effectiveness. The outcomes will enhance understanding and support the development of more effective detection strategies for short-form, dynamic social media platforms such as X.

4. Methodology

4.1. Research design

This study adopts a benchmarking and tool evaluation research design, enriched by qualitative insights gathered through expert interviews, to systematically analyze and compare the effectiveness of widely used AI detection tools. The benchmarking approach provides an objective, structured, and quantitative method to evaluate each tool's accuracy, precision, and reliability in distinguishing between AI-generated and authentic human-generated tweets. The choice of this design aligns effectively with the research objectives, offering several important advantages.

The rapid and ongoing advancements of generative AI models require a methodology capable of robust and nuanced evaluation across diverse textual characteristics. The benchmarking design accommodates this complexity effectively, enabling systematic assessments of detection performance across datasets varying significantly in fluency, linguistic complexity, style, and contextual richness. Such comprehensive benchmarking outcomes are essential for understanding each detection tool's adaptability and robustness in response to continually evolving generative AI technologies. The hypotheses developed in Chapter 3 will be systematically tested through the following structured approach (see Table 1):

Table 1: the developed hypotheses

Hypothesis	Independent Variable (IV)	Dependent Variable (DV)	Testing Approach	Example Interpretation of Results
H1	Sophistication of AI-generated content	Detection methodology complexity	Qualitative & quantitative assessment of which methodologies (simple vs advanced) perform effectively at different content sophistication levels. For instance, comparing performance differences across GPT-2, GPT-3.5, and GPT-4o datasets.	If more advanced datasets (e.g., GPT-4.5) require advanced methods (hybrid, deep learning) for good performance, H1 is supported.
H2	Detection methodology complexity	Effectiveness of AI detection tools	Direct quantitative benchmarking: comparing accuracy, precision, recall, false positives, false negatives among Grammarly (hybrid), ZeroGPT (hybrid), GPTZero (transformer-based deep learning)	If advanced methodologies (e.g., deep learning) show consistently better performance metrics compared to simpler methods, H2 is supported.

Complementing this structured quantitative benchmarking, qualitative expert interviews will add depth and practical relevance to the findings. Interviews conducted with AI researchers and domain experts will provide critical insights into the practical implications, ethical considerations, and methodological challenges associated with current AI-text detection methods. These expert perspectives contextualize the quantitative performance metrics derived from benchmarking, offering explanations and strategic recommendations for future developments and improvements in detection capabilities.

Integrating quantitative and qualitative methods ensures academic rigor while maintaining practical relevance. This dual approach not only generates statistically significant and reliable results but also provides practical insights directly applicable to real-world scenarios. Consequently, the findings

derived from this research will support informed decision-making by platform moderators, policy makers, AI tool developers, and researchers committed to effectively managing and regulating AI-generated content on social media platforms like X. In the following research sequence diagram (Figure 2) will be the methodological approach clearly.

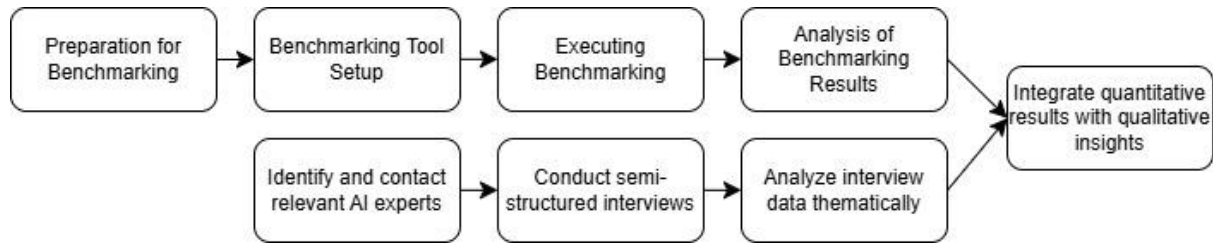


Figure 2: Research sequence diagram

4.2. Selection of AI detection tools

The selection of AI detection tools followed a rigorous two-phase approach, guided by both quantitative data and strategic criteria to ensure methodological rigor and practical applicability. Initially, a comprehensive assessment of the 15 most-visited AI detection websites was conducted using recent web analytics data provided by SimilarWeb (2025). This assessment aimed to objectively identify the most prominent and frequently utilized detection tools globally. The top 15 AI detection tools by total visits from December 2024 to February 2025 are listed below (Table 2), with the complete SimilarWeb report provided in the attached document (see Appendix for complete SimilarWeb analytics).

Table 2: Top 15 AI Detection Tools (Dec 2024 – Feb 2025)

Rank	AI Detection Tool Domain	Total Visits (Dec 2024 - Feb 2025)	Average Visits	Monthly
1	QuillBot (quillbot.com)	171,777,122	57,259,041	
2	Grammarly (grammarly.com)	165,631,544	55,210,514	
3	Scribbr (scribbr.com)	33,310,562	11,103,521	
4	ZeroGPT (zerogpt.com)	45,853,804	15,284,601	
5	GPTZero (gptzero.me)	30,198,921	10,066,307	
6	Undetectable.ai	14,346,351	4,782,117	
7	Originality.ai	4,864,288	1,621,429	
8	WriteHuman.ai	6,435,189	2,145,063	
9	Sapling.ai	2,966,047	988,682	
10	Humbot.ai	2,954,351	984,784	
11	Writer.com	3,190,353	1,063,451	
12	CopyLeaks.com	8,116,651	2,705,550	
13	GoWinston.ai	1,491,487	497,162	

14	Detecting-ai.com	1,085,074	361,691
15	ContentAtScale.ai	73,196	24,399

Following this initial assessment, a structured decision matrix was applied to refine the selection of tools based on specific criteria including suitability for short-form text, API accessibility, transparency of methodologies, industry recognition, and user base. Despite their popularity, QuillBot and Scribbr were excluded from the benchmarking study due to their technical requirement of a minimum of 80 words per sample, incompatible with the brevity of tweets. Given that Scribbr employs QuillBot's detection technology, both were removed from further consideration.

The final tools selected for benchmarking within this study were Grammarly, ZeroGPT, and GPTZero. These tools collectively represent state-of-the-art commercial technologies widely recognized for their effectiveness and practical relevance in AI-generated content detection. Additionally, their methodological diversity ensures comprehensive coverage and comparative insights. A summary of their detection methodologies and features is provided in Table 3 below:

Table 3: Detection Methodologies of Selected AI Tools

Tool	Detection Methodology and Features	Detection Approach
Grammarly	Grammarly employs transformer-based deep learning models that analyze linguistic features, contextual coherence, and syntactic patterns indicative of AI-generated content. It combines semantic embedding analysis with extensive pattern recognition trained on large annotated datasets, enabling it to effectively differentiate between human and AI-generated texts, even in short-form content (Grammarly, 2025).	Hybrid (Feature-Based + Machine Learning)
ZeroGPT	ZeroGPT utilizes advanced transformer-based algorithms specifically optimized to detect content generated by GPT models. It focuses on statistical anomalies, linguistic consistency, and coherence patterns characteristic of GPT-generated text. The tool is regularly updated to adapt to new generative techniques, providing robustness against recent AI advancements (Zerogpt, z.d.).	Deep Learning (Transformer-based)
GPTZero	GPTZero applies a hybrid methodology, integrating statistical measures (such as burstiness and perplexity scores) with transformer-based linguistic pattern analysis. It emphasizes transparency and provides explicit justifications for its AI-text classification decisions, making it widely trusted in academia and journalism. Its model continuously incorporates user feedback and new training data, enhancing adaptability (gptzero.me, z.d.).	Hybrid (Feature-Based + Deep Learning)

Benchmarking these selected tools effectively represents benchmarking the most widely adopted approaches for detecting AI-generated text. Each tool utilizes a distinct methodological combination, spanning from hybrid approaches integrating feature-based analysis with either machine learning or deep learning methods to exclusively transformer-based deep learning. Thus, this benchmark provides a comprehensive evaluation of current, prevalent detection methodologies within the field.

4.3. Benchmark dataset

The benchmarking procedure requires distinct datasets containing diverse textual variations and different generative AI capabilities to ensure a comprehensive evaluation of AI detection tools. In this research, four AI-generated tweet datasets were constructed, alongside a human-generated tweet dataset sourced from Kaggle, originally used in the well-known "TweepFake" study by Tesconi (2023). Table 4 provides an overview of the datasets used in this study, their characteristics, and the justification for their inclusion.

Table 4: Dataset Characteristics

Dataset	Description & Justification
GPT-2	Baseline generative model with moderate fluency. This dataset serves as a reference for early-generation AI text. Due to the older model's limitations, significant filtering was necessary to remove tweets with repetitive patterns, random symbols or irrelevant links.
GPT-3.5	More advanced in fluency and coherence, representing commonly used generative AI. A minor degree of filtering was required to remove some noisy samples, random hashtags, or non-sensical text segments.
GPT-4o	Latest generative model from OpenAI, demonstrating highly sophisticated, near-human-level text generation. The dataset represents cutting-edge capabilities of generative models for producing realistic tweets.
GPT-4o with Emojis	AI-generated tweets with embedded emojis to simulate more realistic and informal social media interactions typical on platforms like X.
Human-generated (TweepFake)	Sourced from real Twitter users via the Kaggle TweepFake dataset. This dataset required filtering to remove tweets consisting solely of links, fragmented sentences, or irrelevant content not aligned with human-authored tweet conventions.

For each AI-generated dataset (GPT-2, GPT-3.5, GPT-4o, and GPT-4o with Emojis), a total of 200 tweets were selected. For the human-generated dataset, 800 tweets were selected. The final dataset size was thus balanced at 1,600 tweets: 800 AI-generated and 800 human-generated.

Initially, a larger collection of tweets was generated or gathered for each category. However, a filtering process was necessary to ensure the quality and authenticity of the data. The GPT-2 dataset, in particular, required extensive cleaning due to its tendency to generate irrelevant content, including random characters (e.g., "@@!!##") or meaningless links (e.g., "<https://123xyz.net>"). Similarly, GPT-3.5 occasionally produced outlier tweets that contained isolated phrases or incomplete thoughts without context, which were removed during the preprocessing phase. The human-generated tweets from Kaggle's TweepFake dataset were also filtered to eliminate tweets that contained only hyperlinks, standalone hashtags, or incomplete conversational fragments that lacked analytical value.

Additionally, for all datasets, tweets shorter than 250 tokens were excluded to maintain a consistent text length for analysis, as prior research suggests detection accuracy tends to decrease with extremely short texts. This rigorous dataset preparation ensures a robust, balanced, and representative set of tweets for benchmarking AI detection tools under real-world conditions.

4.4. Benchmarking procedure

The benchmarking procedure was designed to systematically evaluate the effectiveness of selected AI detection tools in classifying AI-generated versus human-generated tweets. To manage the large dataset

of 1,600 tweets efficiently, the process combined automation technologies with structured evaluation techniques to ensure consistency, accuracy, and replicability.

Given the considerable size of the dataset and the manual limitations of testing each tweet individually, Microsoft Power Automate was utilized to automate the benchmarking process. Power Automate enabled the automated submission of tweets to the online detection platforms of the selected tools: Grammarly, ZeroGPT, and GPTZero. These tools were identified in the selection phase as the most widely used and suitable for short-form text detection on platforms like X.

Each tweet was automatically processed through the respective detection tools' web interfaces. The output of the detection tools indicating whether the tweet was classified as AI-generated or human-written was automatically extracted and stored for further analysis. To ensure full-scale benchmarking without hitting usage caps or access limitations, a premium subscription to GPTZero was purchased, as the free version imposes daily limits that would have restricted the dataset's complete evaluation. Upon completion of the benchmarking process, all results were saved in a structured CSV file, containing the original tweet, the tool's prediction, and additional metadata (such as the model used for generation). This structured data output facilitated the next stage of analysis.

Subsequently, Python programming was employed to analyze the benchmarking results. Using Python's data analysis libraries (such as pandas and scikit-learn), the dataset was processed to calculate key performance metrics for each detection tool. These metrics are consistent with established academic standards for evaluating classification models and were further supported by expert interviews conducted in this research. The key evaluation metrics used in this study are presented in Table 5:

Table 5: Effectiveness Metrics

Metric	Description
Accuracy	Proportion of correctly classified tweets overall.
Precision	Percentage of tweets classified as AI-generated that were correctly identified.
Recall (Sensitivity)	Percentage of actual AI-generated tweets correctly identified by the tool.
False Positives	Rate of human-generated tweets incorrectly classified as AI-generated.
False Negatives	Rate of AI-generated tweets incorrectly classified as human-generated.

These evaluation metrics provide comprehensive insights into the strengths and limitations of each detection tool. They allow for the assessment of potential biases, error tendencies, and overall robustness of the tools in handling diverse AI-generated and human-generated content on social media platforms like X.

The combined use of Power Automate for large-scale data collection, along with Python for statistical analysis, ensures methodological rigor while maintaining practical feasibility. This approach provides a replicable framework for future benchmarking studies in the field of AI-generated content detection.

4.5. Calculation Methods

To evaluate the classification performance of the AI detection tools, a set of standard binary classification metrics were used: accuracy, precision, recall (sensitivity), false positives, and false negatives. This section defines each metric and explains how it was calculated from the benchmarking results using Python. All calculations were based on the comparison between actual labels (ground truth: AI or human) and predicted labels (outputs from the detection tools). The terms in table 6 are used in the metric formulas.

Table 6: Terms Definitions

Abbreviation	Term	Description
TP	True Positive	Number of AI-generated tweets correctly classified as AI
TN	True Negative	Number of human-generated tweets correctly classified as human
FP	False Positive	Number of human-generated tweets incorrectly classified as AI
FN	False Negative	Number of AI-generated tweets incorrectly classified as human

Accuracy

Accuracy quantifies the overall correctness of a detection tool by calculating the proportion of all tweets correctly identified, whether AI-generated or human-generated. The formula of accuracy is:

$$Accuracy = \frac{TP + TN}{FP + FN + TP + TN}$$

A high accuracy suggests general reliability. However, accuracy alone may be misleading in imbalanced datasets, as a high accuracy score could mask weaknesses in accurately detecting the minority class.

Precision

Precision specifically assesses the accuracy of predictions labeled as AI-generated. It indicates how many of the tweets identified by the tool as AI-generated were genuinely produced by AI. The formula of precision is:

$$Precision = \frac{TP}{FP + TP}$$

A high precision value indicates fewer false alarms (false positives). Precision is crucial when falsely identifying human-generated content as AI-generated could have significant consequences, particularly in sensitive fields like journalism or academia.

Recall (Sensitivity)

Recall, also known as sensitivity, measures the tool's effectiveness in correctly identifying all tweets that are genuinely AI-generated. It reveals the proportion of AI-generated tweets accurately detected. The formula of recall is:

$$Recall = \frac{TP}{TP + FN}$$

A high recall indicates that most AI-generated content is successfully identified. Maximizing recall is particularly important in contexts where allowing AI-generated misinformation to pass unnoticed poses substantial risks.

False Positives Rate (FPR)

The False Positive Rate measures how frequently the tool incorrectly classifies human-generated tweets as AI-generated, representing the proportion of genuine human tweets misidentified. The formula of FPR is:

$$False\ Positive\ Rate = \frac{FP}{FP + TN}$$

A high false positive rate may reduce user trust and satisfaction, as legitimate content could be mistakenly flagged, censored, or penalized.

False Negatives Rate (FNR)

The False Negative Rate indicates how frequently AI-generated tweets are mistakenly classified as human-generated, thus escaping detection. The formula of FNR is:

$$\text{False Negative Rate} = \frac{FN}{FN + TP}$$

A high false negative rate signifies that the detection tool frequently fails to identify AI-generated content, potentially enabling misinformation or synthetic content to disseminate undetected.

The benchmarking results are systematically analyzed using Python’s data analysis libraries, including pandas and scikit-learn. For each of the evaluated detection tools (Grammarly, ZeroGPT, and GPTZero) a confusion matrix is generated to obtain the values of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These values are applied to the defined metric formulas to determine the performance metrics for each detection tool.

4.6. Expert Interviews

To supplement quantitative findings, qualitative insights were gathered through semi-structured expert interviews. Five AI researchers were selected based on their expertise in generative AI, NLP, and AI ethics, ensuring a comprehensive understanding of the field. Interviews covered their perspectives on current AI detection effectiveness, tool biases, and potential enhancements, guided by carefully designed questions as outlined below (Table 7):

Table 7: Interview Questions Overview

Interview Segment	Questions Asked
General Perceptions	Opinions on current AI detection tools and general reliability.
Technical and Ethical Considerations	Discussions on accuracy, biases, and the implications of misclassification on social media platforms.
Future Improvements and Predictions	Recommendations for improving AI detection accuracy and foresight on technological evolution over the next 5-10 years.
Additional Insights	Open-ended invitation for further comments and recommendations relevant to the research.

The expert interviews provided contextually rich insights, enhancing understanding of real-world implications and informing recommendations for tool improvement.

4.7. Ethical Considerations and Reliability Measures

This research was conducted with strict adherence to ethical and academic standards, ensuring both the protection of participants and the validity of the data collected.

4.7.1. Ethical Compliance

To maintain ethical integrity, all tweet datasets used in this study were either synthetically generated using generative AI models (GPT-2, GPT-3.5, GPT-4o) or obtained from publicly available datasets such as the *TweepFake* dataset (Tesconi, 2023). No private or sensitive user data was collected, and no tweets were sourced in a way that could violate user confidentiality or platform terms of service.

For the expert interviews, informed consent was obtained from all participants prior to participation. Each expert was fully briefed on the purpose of the study, the nature of the questions, and how the data would be used. Participants were given the option to remain anonymous, and all responses were stored

securely with personal identifiers removed to maintain privacy. The interview process was approved under academic guidelines for ethical research involving human subjects.

4.7.2. Reliability Measures

To ensure reliability and consistency of results, several methodological steps were implemented across both the benchmarking and interview phases of the study:

- **Benchmarking Replication:** Each tweet was processed twice through the detection tools (Grammarly, GPTZero, ZeroGPT) to assess consistency in the output. This repeated testing aimed to identify any discrepancies or random variations in detection outcomes. The results showed high consistency across all tools, reinforcing the reliability of the collected data.
- **Controlled Automation:** Microsoft Power Automate was used to process the entire set of 1,600 tweets in a controlled and standardized environment. This minimized human error and ensured uniform handling of inputs across all detection tools.
- **Consistent Evaluation Metrics:** The same evaluation metrics (accuracy, precision, recall, false positives, and false negatives) were applied to all tools using a single Python script. This ensured consistent data processing and reduced potential variability in how performance was assessed.
- **Structured Interviews:** A single set of structured, pre-validated questions was used for all five expert interviews. This standardization allowed for more reliable comparison and synthesis of qualitative insights across respondents. Interviews were recorded (with permission), transcribed, and analyzed systematically.
- **Cross-validation of Interpretation:** During data analysis and hypothesis testing, results were triangulated across different datasets (GPT-2 to GPT-4o) and detection tools. Where results diverged or were unexpected, they were revisited and interpreted with caution, referencing both expert input and supporting literature.

5. Results

This chapter presents the empirical findings derived from the benchmarking evaluation of selected AI-generated text detection tools (Grammarly, GPTZero, ZeroGPT) on datasets with varying degrees of generative AI sophistication (GPT-2, GPT-3.5, GPT-4o, GPT-4o with emojis) and human-generated tweets. These findings provide insights into the effectiveness of different detection methodologies in identifying AI-generated content on social media platforms like X (formerly Twitter). The results are presented systematically, first at the dataset level, and subsequently at the tool level, supported by clear metrics and visualizations.

5.1. Overall Performance by Dataset

The performance metrics across datasets reveal notable differences in detection accuracy, precision, and recall, depending significantly on the generative sophistication of the datasets. The performance outcomes for each dataset are summarized in Table 8.

Table 8: Performance Metrics by Dataset

Dataset	Accuracy	Precision	Recall
GPT-2	0.8867	0.7167	0.2233
GPT-3.5	0.9250	0.9600	0.7800
GPT-4o	0.9775	0.9967	0.9700
GPT-4o with Emojis	0.8233	0.9133	0.6467
Human	0.9750	0.0000	0.0000

These results were derived from a total of 1,600 tweets: 200 tweets from each AI-generated dataset and 800 human-generated tweets. Each tweet was evaluated using three AI detection tools and the predictions were collected using Microsoft Power Automate. These predictions were stored and analyzed using Python, where confusion matrices were constructed for each dataset. From these, the values of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) were extracted and used to calculate the metrics based on the formulas in chapter 4. By averaging the scores across the three tools for each dataset, the final performance values were computed. The benchmarking results yield unexpected insights regarding the relationship hypothesized in H1. Contrary to the original expectation that increasingly sophisticated generative AI content would require progressively more complex detection methods (thus becoming harder to detect), the empirical data reveals an inverse pattern. Specifically, the GPT-4o dataset demonstrated significantly higher recall (97%) and overall detection accuracy across all tested tools. In contrast, the GPT-2 dataset exhibited the lowest recall (22.33%), indicating substantial difficulties in classification, despite being the least sophisticated generative model.

These findings indicate that more sophisticated generative AI models, like GPT-4o, may produce distinctive linguistic features or patterns that advanced detection methodologies (such as transformer-based models employed by GPTZero) are particularly adept at identifying. Conversely, simpler AI-generated content (e.g., GPT-2) may produce less coherent, more randomized outputs, making them harder to distinguish from low-quality human-generated text.

GPT-3.5, positioned between GPT-2 and GPT-4o in generative sophistication, exhibited strong performance with an accuracy of 92.5%, a precision of 96%, and a recall of 78%. The dataset combining GPT-4o text with emojis produced an accuracy of 82.33% and a recall of 64.67%, slightly lower than GPT-4o without emojis. This implies that informal textual features such as emojis may dilute or obscure the patterns that detection tools rely on, thereby lowering their effectiveness.

The human-generated dataset, consisting solely of genuine tweets, recorded a high accuracy score of 97.5%. However, precision and recall were both at 0%, as expected, since the dataset contained no actual AI-generated tweets. This outcome highlights the importance of evaluating false positives, which can have significant implications in practical applications.

These findings are visualized in Figure 3, which compares detection performance across the datasets. The chart clearly illustrates the contrast in detection success between the most and least sophisticated generative models, as well as the effects of stylistic variations such as emojis.

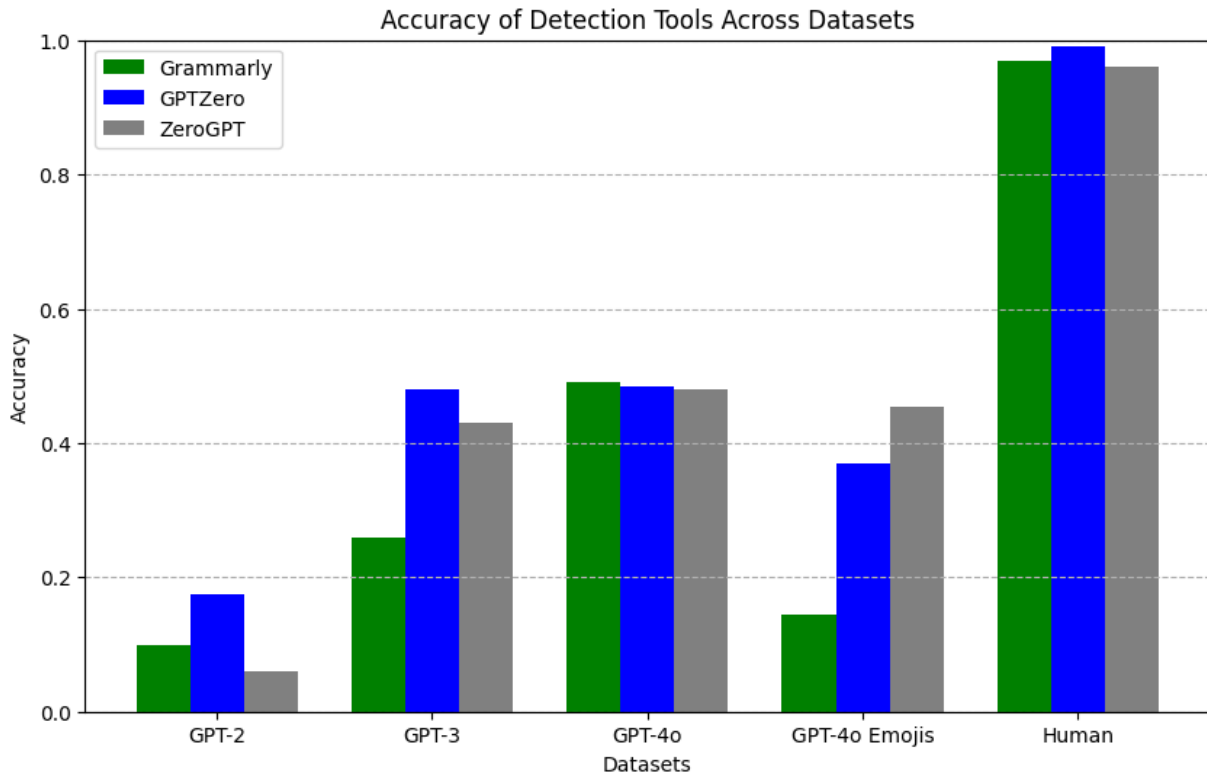


Figure 3: Accuracy Results of AI Detection Tools Across Datasets

The data suggests that more sophisticated generative models are not necessarily harder to detect. In fact, GPT-4o was the easiest for tools to classify correctly, while GPT-2 posed the greatest challenge. Therefore, these findings do not support Hypothesis 1 (H1) as initially stated. Instead, they suggest an alternative interpretation that the relationship between generative AI sophistication and detection difficulty is not linear or direct. Rather, detection effectiveness might depend more on specific linguistic and stylistic markers associated with generative model complexity.

5.2. Overall Performance by Detection Tool

The benchmarking procedure further assessed the performance of the selected detection tools across all combined datasets, focusing on accuracy, precision, and recall metrics. Table 9 summarizes these overall findings:

Table 9: Performance Metrics by Detection Tool

Detection Tool	Accuracy	Precision	Recall
Grammarly	0.7338	0.9431	0.4975
GPTZero	0.8725	0.9869	0.7550
ZeroGPT	0.8363	0.9468	0.7125

Figure 4 illustrates the comparative performance of the detection tools. GPTZero demonstrated superior performance, achieving the highest overall accuracy (87.25%), precision (98.69%), and recall (75.50%), followed closely by ZeroGPT. Grammarly, while maintaining high precision (94.31%), demonstrated notably lower recall (49.75%) and accuracy (73.38%).

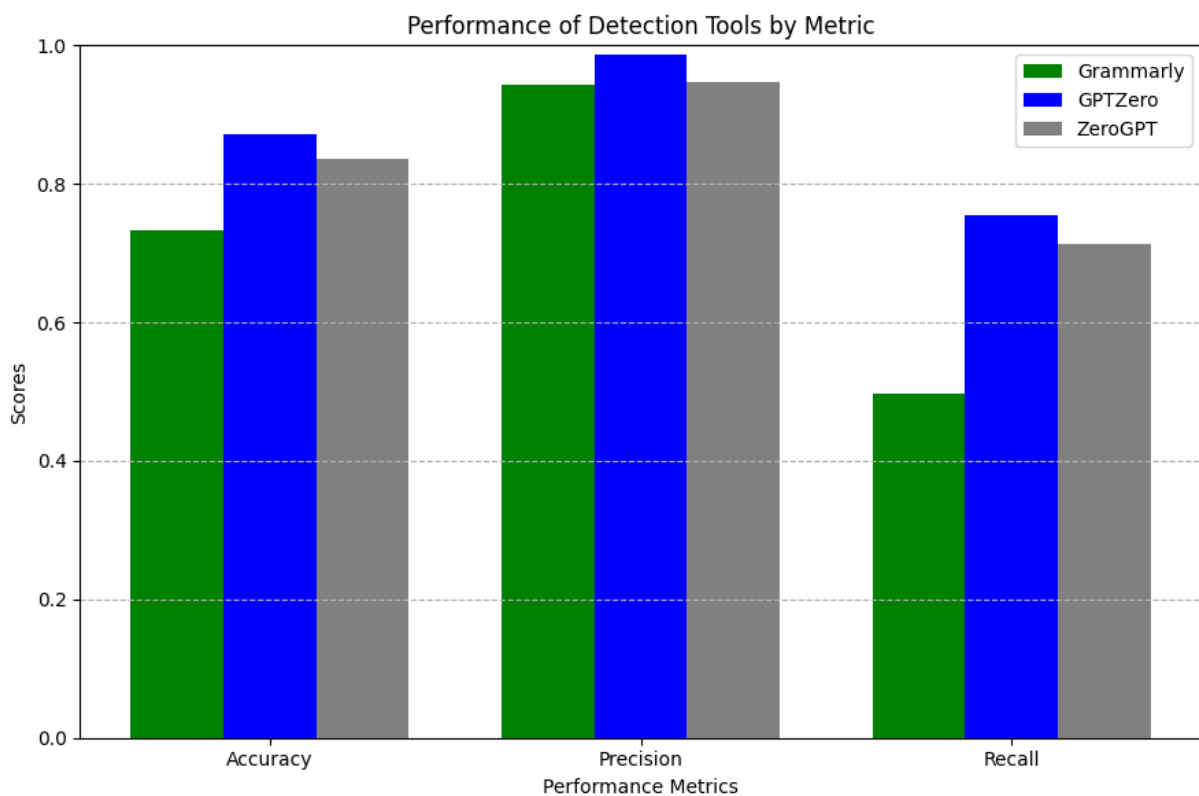


Figure 4: Performance of Each AI Detection Tool by Metric

These outcomes provide empirical support for Hypothesis 2 (H2), suggesting that more complex detection methodologies (such as GPTZero’s transformer-based deep learning approach) are indeed positively correlated with improved effectiveness in identifying AI-generated content. GPTZero employs a multi-faceted approach that includes deep learning components and statistical analyses. Specifically, it utilizes metrics such as perplexity and burstiness to assess the randomness and variability of text, which are indicative of human versus AI authorship. Additionally, GPTZero incorporates seven machine learning components, including an Education Module and Internet Text Search, to enhance its detection capabilities (gptzero.me, z.d.).

ZeroGPT adopts a hybrid methodology that combines deep learning algorithms with natural language processing techniques. Its proprietary DeepAnalyse™ technology conducts a multi-stage analysis, examining both macro-level structures and micro-level linguistic features to determine the likelihood of AI authorship (Zerogpt, z.d.).

In contrast, Grammarly relies primarily on statistical models trained on datasets comprising human-written and AI-generated texts. Its detection mechanism analyzes language patterns commonly associated with AI-generated content but does not incorporate deep learning techniques to the same extent as GPTZero or ZeroGPT. This reliance on statistical analysis may limit its ability to detect more sophisticated AI-generated texts (Grammarly, 2025).

In summary, the advanced methodologies employed by GPTZero and ZeroGPT, which integrate deep learning and comprehensive linguistic analyses, contribute to their higher effectiveness in detecting AI-generated content. These results substantiate Hypothesis 2, affirming that the complexity and adaptability of a detection tool's methodology are critical factors in its performance.

5.3. Detailed Analysis of Classification Errors

A detailed breakdown of false positives and false negatives provides additional insights into detection performance limitations and potential biases inherent in each tool. Table 10 summarizes these error rates.

Table 10: Error Rates

Detection Tool	False Positive Rate (%)	False Negative Rate (%)
Grammarly	3.00%	50.25%
GPTZero	1.00%	24.50%
ZeroGPT	4.00%	28.75%

Grammarly and ZeroGPT exhibited higher false negative rates, suggesting they occasionally fail to detect sophisticated AI-generated tweets, while GPTZero maintained lower error rates in both categories. This underscores GPTZero's robustness and aligns with its reliance on advanced deep learning methodologies, which have proven more adaptable to nuanced and sophisticated content.

5.4. Statistical Testing of Detection Tool Performance

To further validate the observed performance differences between the detection tools, independent-sample t-tests were conducted. The tests compared the overall detection outcomes (expressed in raw detection percentages) across Grammarly, GPTZero, and ZeroGPT. The goal was to statistically assess whether the performance differences among the tools were significant or simply due to random variation. The t-tests were conducted on the raw output percentages collected from the benchmarking procedure. For each comparison, the null hypothesis assumed no difference in detection performance between the tools, while the alternative hypothesis assumed a significant difference. The results are summarized in Table 11:

Table 11: Statistical Results

Comparison	t-statistic	p-value	Interpretation
Grammarly vs GPTZero	-7.461	1.11×10^{-13}	Significant difference ($p < 0.001$)
Grammarly vs ZeroGPT	-6.526	7.81×10^{-11}	Significant difference ($p < 0.001$)
GPTZero vs ZeroGPT	1.077	0.281	No significant difference ($p > 0.05$)

The statistical results reveal that both GPTZero and ZeroGPT significantly outperformed Grammarly in detecting AI-generated content. Specifically, the p-values for comparisons between Grammarly and the other two tools were well below the conventional threshold of 0.05, confirming a statistically significant difference. Thus, Grammarly demonstrated weaker overall performance relative to GPTZero

and ZeroGPT. In contrast, the comparison between GPTZero and ZeroGPT yielded a non-significant p-value ($p = 0.281$), indicating no meaningful statistical difference in their detection capabilities. Although GPTZero had a slightly higher observed mean performance, the difference was not strong enough to conclude a superiority over ZeroGPT at the 95% confidence level.

These findings strengthen the support for Hypothesis 2 (H2), which proposed that detection tools employing more advanced methodologies, such as transformer-based deep learning (GPTZero), would exhibit superior performance. The statistical tests corroborate that Grammarly's relatively simpler heuristic-based detection approach was consistently outperformed by the more advanced models. Overall, the statistical analysis provides robust evidence that methodological complexity is positively associated with detection performance, reinforcing the benchmark results observed earlier.

5.5. Interviews findings

To complement the quantitative benchmarking analysis, semi-structured expert interviews were conducted with five renowned AI researchers. The objective was to gain comprehensive qualitative insights into the current effectiveness, limitations, ethical considerations, and future potential of AI-generated content detection on social media platforms. Four experts (Professors G. Cascavilla, H. Brouwer, G. Chrupała, and M. Alimardini) agreed to be identified, while a fifth interviewee preferred to remain anonymous. The findings from these interviews are structured into four thematic categories: General Perceptions, Technical and Ethical Considerations, Future Improvements and Predictions, and Additional Insights.

5.5.1. General Perceptions on AI Detection Tools

Overall, all interviewed experts expressed significant skepticism regarding the effectiveness and reliability of current AI detection tools. Professor Harm Brouwer succinctly captured this sentiment, describing existing detection tools as "not reliable enough for practical use." He explained that AI-generated content often resembles what he termed "hollow" text linguistically sophisticated yet lacking substantive depth, making consistent detection particularly challenging.

Professor G. Cascavilla reinforced this viewpoint by criticizing the reliance on conventional detection methodologies such as basic machine learning classifiers trained on static datasets of human and AI-generated texts. He labeled such approaches as "fundamentally flawed" and "unreliable at scale," particularly given the rapid evolution of generative AI technologies.

Professor G. Chrupała characterized the current situation as an "arms race," highlighting a perpetual cycle of incremental advancements in generative models and responsive improvements in detection methodologies. This dynamic ensures that detection tools constantly lag behind state-of-the-art generative techniques, undermining their long-term effectiveness.

Similarly, the anonymous researcher provided a critical perspective, emphasizing that the sophistication of generative AI is rapidly increasing, rendering current detection strategies increasingly obsolete. According to this expert, detection tools inevitably face diminishing returns due to the continuous improvement of generative AI.

5.5.2. Technical and Ethical Considerations

Interviewees unanimously emphasized critical concerns surrounding the technical limitations and ethical implications associated with AI detection technologies. Professor Cascavilla discussed the significant technical challenges of adapting detection models to social media platforms like X. Although technically possible, he argued that training effective detection systems on real-time social media data would require immense computational resources, extensive manual annotation, and careful handling of privacy consideration factors that significantly limit feasibility.

Professor Brouwer introduced an additional critical consideration: the impact of self-referential training data. He noted that as generative models increasingly train on internet data which contains AI-generated

content, there emerges a problematic feedback loop. Such loops potentially lead to biased outcomes and a homogenization of generated content, complicating detection by reducing clear distinctions between human-authored and AI-generated texts.

The ethical dimension of false classifications, especially false positives, was identified as a major concern by multiple experts. Professor Alimardini emphasized the consequences of mistakenly labeling human-authored content as AI-generated. Such misclassifications could damage credibility, stifle legitimate discourse, or even lead to censorship, particularly in sensitive contexts such as journalism, academia, or political communication.

5.5.3. Future Improvements and Predictions

Interviewees offered thoughtful, albeit cautious, predictions about the future trajectory of AI-generated content detection. Importantly, none of the experts expressed strong optimism about the eventual emergence of a universally effective detection mechanism. Professor Cascavilla explicitly voiced his doubts about the future of robust AI detection tools, asserting that generative AI models like GPT series are improving at a much faster pace than detection capabilities. This disparity, he argued, significantly limits the likelihood of achieving long-term effective detection.

Professor Brouwer, alongside the anonymous researcher, echoed this sentiment by emphasizing that detection tools are persistently reactive rather than proactive. Both suggested a potential need for paradigm shifts in detection technology moving beyond conventional statistical or syntactic analyses to more complex semantic and contextual understanding. However, they acknowledged that such shifts face considerable obstacles, given the current state of natural language processing and computational linguistics.

Professor Alimardini added that integrating non-textual data such as metadata, user behavior patterns, and historical interaction data might offer incremental improvements. Nonetheless, she cautioned that such measures might pose further ethical and privacy concerns, potentially limiting practical implementation.

5.5.4. Additional Insights and Broader Implications

The interviews also illuminated broader reflections on the societal and cultural implications of widespread AI-generated text. Professor Brouwer offered a sociocultural perspective, arguing that writing is fundamentally a social practice deeply tied to human intention, context, and interaction. He suggested that the increased reliance on AI-generated text could fundamentally alter the nature of written communication, reducing it to algorithmically produced content disconnected from genuine human intent. Several interviewees pointed to the growing ubiquity of generative AI tools among students, professionals, and everyday users. Professor Chrupała specifically highlighted how the normalization of AI-assisted writing, particularly through platforms such as ChatGPT is blurring the boundaries between human and AI-generated texts. This blurring complicates the moderation and authenticity of content on platforms like X, raising significant questions about credibility and accountability.

The anonymous expert offered a particularly sobering insight: "It seems unlikely we will ever reach a stage where AI-generated content can be reliably and consistently detected." This remark captures the collective caution and uncertainty prevalent among the interviewed experts, highlighting the complexity, uncertainty, and ongoing challenges associated with moderating AI-generated content.

5.5.5. Summary of Interview Insights

Overall, the expert interviews reinforced the quantitative benchmarking findings by emphasizing several key points:

- Current detection tools, although moderately effective against certain AI-generated text models, lack robustness and long-term reliability.
- Ethical, privacy, and societal implications must be central considerations in developing and deploying AI detection methodologies.
- Future research must explore more sophisticated semantic analysis and hybrid methodologies, despite considerable technical and resource-related challenges.
- The rapid evolution of generative AI models, combined with the growing ubiquity of AI-generated content, requires continuous adaptation and realistic expectations about detection capabilities.

Collectively, these insights underscore the complex, evolving nature of AI detection on social media platforms, highlighting both the practical limitations and critical need for ongoing research and innovation in this rapidly changing domain.

6. Discussion and Conclusion

6.1. Discussion

The objective of this study was to evaluate the effectiveness of current AI-generated text detection tools in the context of social media platforms, particularly focusing on X (formerly Twitter). By leveraging both benchmarking methodologies and expert interviews, this research contributes a multidimensional understanding of the capabilities, limitations, and implications of detection systems in the evolving landscape of generative AI.

6.1.1. Insights from Benchmarking: Revisiting Hypotheses

Hypothesis 1 (H1) posited that more sophisticated AI-generated content would necessitate more advanced detection methodologies due to increased difficulty in distinguishing AI-generated text from human-written content. Contrary to this assumption, the results revealed an inverse trend: GPT-4o, the most advanced model tested, was detected with the highest accuracy and recall across all tools, whereas the older GPT-2 model yielded the lowest detection success. These findings suggest that more sophisticated models may leave detectable traces or linguistic consistencies that current detection tools, particularly those based on deep learning or hybrid methods, can exploit effectively.

This challenges the assumption of a linear correlation between AI model sophistication and detection difficulty. Instead, the data implies that detection difficulty may be more strongly influenced by stylistic irregularities or coherence gaps, which are more common in outputs from older, less advanced models. These findings do not support H1 in its original form, indicating that complexity alone does not equate to detection difficulty.

Hypothesis 2 (H2) asserted that detection tools using more complex methodologies, especially those integrating deep learning and hybrid approaches would outperform simpler, statistical models. This hypothesis was strongly supported by both benchmarking results and statistical testing. GPTZero and ZeroGPT, which employ advanced detection methodologies, significantly outperformed Grammarly, which relies primarily on statistical modeling. GPTZero achieved the highest levels of accuracy (87.25%), precision (98.69%), and recall (75.5%). This supports the notion that methodological sophistication is a key driver of detection performance, reinforcing calls in the literature for more adaptive and context-aware detection systems.

Statistical validation using T-tests further confirmed the significance of these performance differences. Grammarly's performance was significantly lower ($p < 0.001$) compared to GPTZero and ZeroGPT, both of which showed no statistically significant difference between them. This finding bolsters the credibility of GPTZero and ZeroGPT as more reliable options for detecting AI-generated content in short-form social media contexts.

6.1.2. Error Patterns and Their Implications

The analysis of classification errors uncovered crucial limitations in the current state of AI detectors. While false positives (misclassifying human tweets as AI) were relatively low across tools, false negatives (failing to detect AI-generated tweets) were markedly higher for tools like Grammarly and ZeroGPT, with false negative rates of 50.25% and 28.75%, respectively. These errors are especially problematic in environments like social media, where the undetected spread of AI-generated misinformation or spam can have significant societal implications.

Furthermore, the performance discrepancy between GPT-4o with and without emojis highlighted another challenge: informal or stylized elements like emojis may obscure linguistic patterns essential for accurate classification. This observation aligns with existing literature which stresses the difficulty of detecting AI-generated text in dynamic and informal online contexts (Cui et al., 2023).

6.1.3. Expert Perspectives: Practical and Ethical Dimensions

The interviews with five AI researchers provided a deeper, more critical perspective on the capabilities and future of AI detection. Across all interviews, a clear theme emerged: skepticism. Experts broadly agreed that existing tools are unreliable and insufficiently robust to meet the challenges posed by generative models. Professor Brouwer noted the “hollow” quality of AI text grammatically correct but semantically shallow, while Dr. Cascavilla emphasized that traditional detection approaches are simply “not doable” at scale.

Technical and ethical limitations were also emphasized. Interviewees expressed concern over tools trained on social media content, citing user privacy issues and the risk of model feedback loops, where AI-generated content is used as training data, further eroding the boundary between human and machine writing. False positives were seen as particularly problematic in high-stakes settings like academia or journalism, where erroneous AI labels could have reputational consequences.

No expert interviewed believed that highly reliable, general-purpose detectors were likely to emerge in the future. Instead, some proposed integrating metadata analysis, behavioral signals, or platform-specific linguistic markers as potential improvements. However, these ideas were acknowledged as conceptually sound but difficult to implement. The prevailing sentiment was that detection systems would continue to lag behind generative models, a dynamic described by several interviewees as an “arms race.”.

6.1.4. Methodological Reflection and Reliability

This study’s methodology, combining automated benchmarking with expert interviews, proved effective for capturing both technical performance and nuanced human perspectives. Microsoft Power Automate provided a scalable mechanism for testing 1,600 tweets across three tools, while Python’s statistical analysis libraries ensured transparent and replicable metric calculations. Benchmarking replication and standardized interview protocols contributed to high methodological reliability.

Nonetheless, limitations remain. The study did not explore multimodal content or detection systems that incorporate behavioral or historical user data. Additionally, as AI-generated content becomes increasingly multimodal (e.g., integrating text, voice, and images), future studies must expand their scope accordingly.

6.2. Conclusion

This study systematically evaluated the effectiveness of widely adopted AI detection tools like Grammarly, GPTZero, and ZeroGPT in identifying AI-generated content on X, a social media platform characterized by short-form, contextually sparse textual communication. By benchmarking 1,600 tweets across different levels of generative AI sophistication (GPT-2, GPT-3.5, GPT-4o, and GPT-4o with emojis) and supplementing these findings with qualitative insights from expert interviews, the study revealed several crucial insights.

Firstly, contrary to the initial hypothesis, the study found that more recent and advanced generative models, such as GPT-4o, were significantly easier to detect compared to earlier models like GPT-2. This result challenges the intuitive assumption that increased generative sophistication inherently corresponds to greater detection difficulty. Instead, the findings suggest that advanced generative models produce text with identifiable linguistic and stylistic signatures more amenable to detection methods that utilize complex analytical approaches, such as transformer-based deep learning or hybrid detection pipelines.

Secondly, the effectiveness of AI detection was found to depend significantly on the underlying detection methodology. The benchmarking clearly demonstrated that tools employing sophisticated methodologies, particularly those that integrate deep learning techniques and hybrid feature-based analysis, consistently outperformed simpler, heuristic-based models. GPTZero and ZeroGPT, both of

which incorporate advanced transformer architectures and sophisticated statistical measures, delivered higher accuracy, precision, and recall than Grammarly, reinforcing the importance of methodological complexity in achieving reliable detection.

The qualitative component of this research, derived from interviews with AI researchers, further contextualized these quantitative findings. Despite advancements in detection methodologies, the experts universally expressed skepticism regarding the possibility of developing universally effective and permanently reliable AI detection tools. They highlighted significant challenges, including privacy concerns, ethical dilemmas associated with misclassification, and the broader societal implications of widespread reliance on AI-generated text. Crucially, experts underscored the dynamic and rapidly accelerating evolution of generative AI technologies, predicting that detection tools, even those currently performing well, will inevitably face increasing difficulties as generative models improve.

In conclusion, while current detection systems have demonstrated the capability to accurately classify texts from certain advanced AI models, it is imperative to recognize that this success is temporary and context dependent. Given the ongoing and rapid evolution of generative AI, today's detection capabilities provide no absolute guarantee for future reliability. As AI-generated text continues to evolve, becoming increasingly indistinguishable from authentic human communication, existing detection methodologies will require constant updating and refinement to remain relevant. Policymakers, platform moderators, and developers must remain vigilant and adaptive, continually integrating new insights and technologies to counteract the ever-improving capabilities of generative AI models.

6.3. Recommendations

It is crucial to adopt hybrid and modular detection architectures that integrate multiple analytical approaches. Detection tools like GPTZero, which combine syntactic analysis, statistical modeling, and transformer-based deep learning, proved to be the most accurate and reliable in this study. Such hybrid systems benefit from the strengths of each method: the interpretability of feature-based analysis, the adaptability of machine learning, and the contextual depth of neural networks. Going forward, modular tool designs that can adapt dynamically to newly emerging generative strategies will be better equipped to cope with the fast-evolving landscape of AI-generated content (Fagni et al., 2021; Kumarage et al., 2023).

There is also an urgent need to enhance the real-time detection capabilities of these tools, particularly when dealing with short-form content like tweets. Benchmarking results showed that some tools struggle with brevity, informal language, or the use of emojis. As prior research highlights, tweets often lack the textual density required for deep semantic models to function optimally (Cui et al., 2023; Ippolito et al., 2019). Detection tools should therefore be retrained on short-form, informal, and platform-specific data and incorporate stylometric and character-level markers such as sentence complexity, punctuation frequency, or emoji distribution to boost accuracy.

Relying solely on textual input is no longer sufficient in today's multimodal, cross-platform digital ecosystem. Developers should begin to incorporate metadata and behavioral signals to improve detection accuracy. Information such as post timing, device usage, geolocation patterns, and historical posting behavior can provide essential context for determining whether a post was AI-generated. For example, research by Rossi et al. (2022) and Yang et al. (2024) has shown that bot-like behavior, when combined with AI-generated text, is a strong indicator of inauthentic activity. Integrating these behavioral insights could significantly reduce false positives and enhance model precision, especially during high-risk periods such as elections or public health crises.

Equally important is the need to promote transparency and explainability in the design and deployment of AI detection systems. Many current tools operate as black boxes, offering minimal insight into why certain content was flagged. This opacity undermines trust, particularly in academic, journalistic, and

legal environments where classification errors could result in censorship or reputational harm. As advocated by Rai (2019), implementing explainable AI (XAI) principles will foster accountability and user confidence. Tools like GPTZero, which already provide justifications for their decisions, should be regarded as exemplary in this regard (Elkhatat et al., 2023).

Lastly, effective AI detection will require cross-sector collaboration and ethical oversight. As generative technologies become more integrated into digital communication, the responsibility for detection cannot fall solely on tool developers or platform moderators. Policymakers, educational institutions, civil society, and private firms must jointly establish detection standards, audit mechanisms, and ethical frameworks for deployment. This aligns with calls from Pan et al. (2024), who stress the importance of harmonized regulatory approaches in managing the dual threats of misinformation and algorithmic overreach. Regulatory frameworks should also address data privacy, model bias, and the potential misuse of detection systems to suppress dissent or amplify surveillance.

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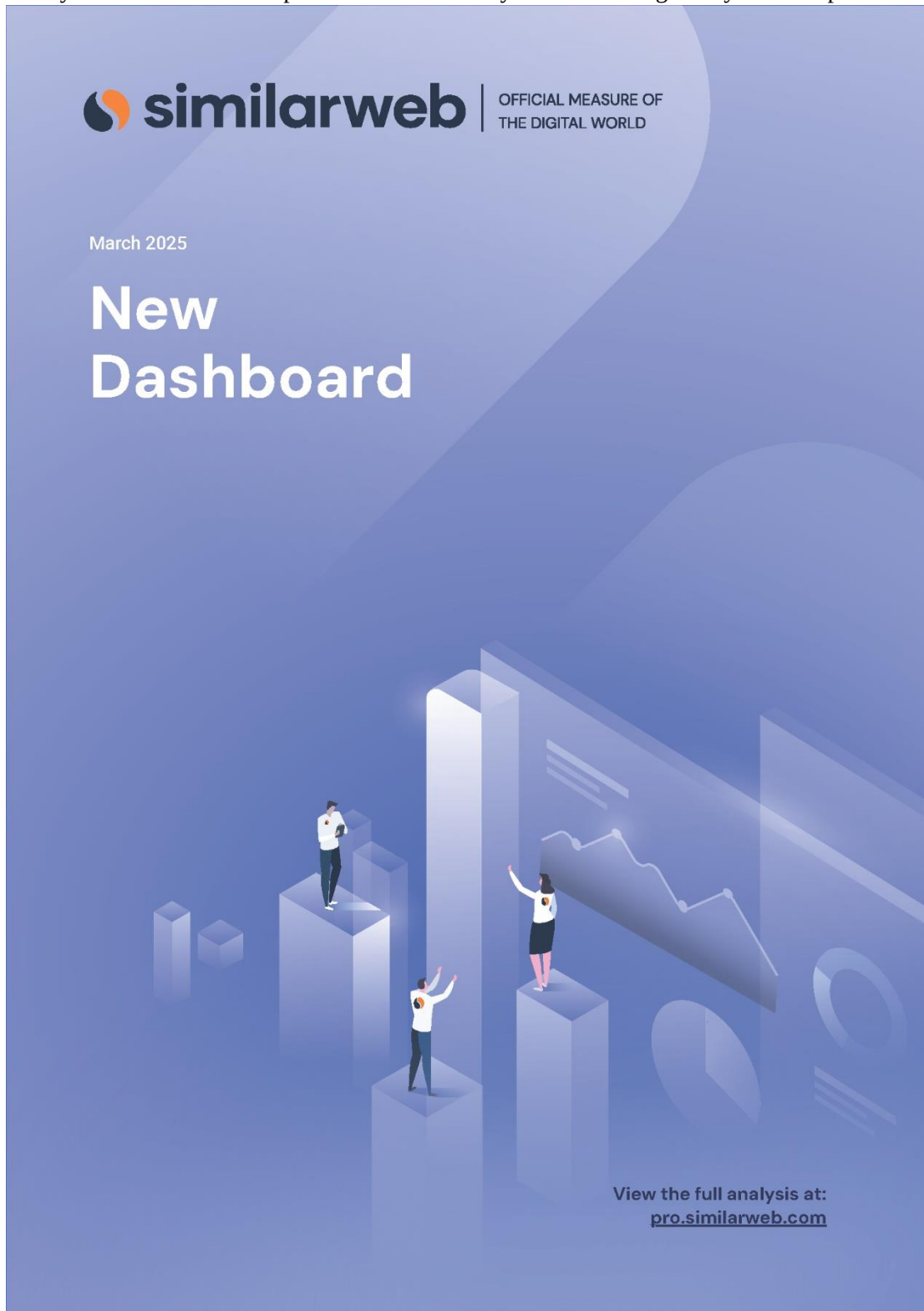
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Appendices

A. Similar Web Analytics report

Here you can find the full report of the most widely used AI tools globally over the past three months.



Total Visits

Dec 2024 - Feb 2025 Worldwide

Domain	Total Visits
originality.ai	4,864,288
zerogpt.com	45,853,804
gptzero.me	30,198,921
writer.com	3,190,353
gowinston.ai	1,491,487

Total Visits

Dec 2024 - Feb 2025 Worldwide



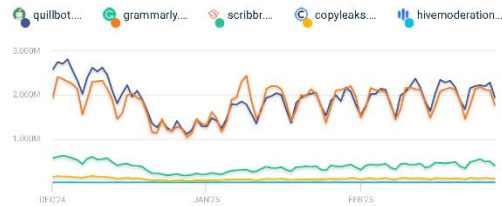
Total Visits

Dec 2024 - Feb 2025 Worldwide

Domain	Total Visits
quillbot.com	171,777,122
grammarly.com	165,631,544
scribbr.com	33,310,562
copyleaks.com	8,116,651
contentatlarge.ai	73,196

Total Visits

Dec 2024 - Feb 2025 Worldwide



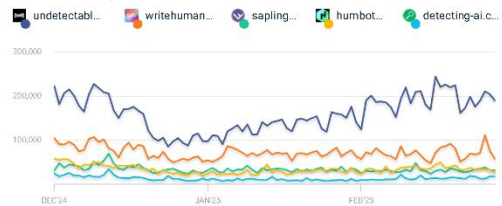
Total Visits

Dec 2024 - Feb 2025 Worldwide

Domain	Total Visits
undetectable.ai	14,346,351
writehuman.ai	6,435,189
sapling.ai	2,966,047
humbot.ai	2,954,351
detecting-ai.com	1,085,074

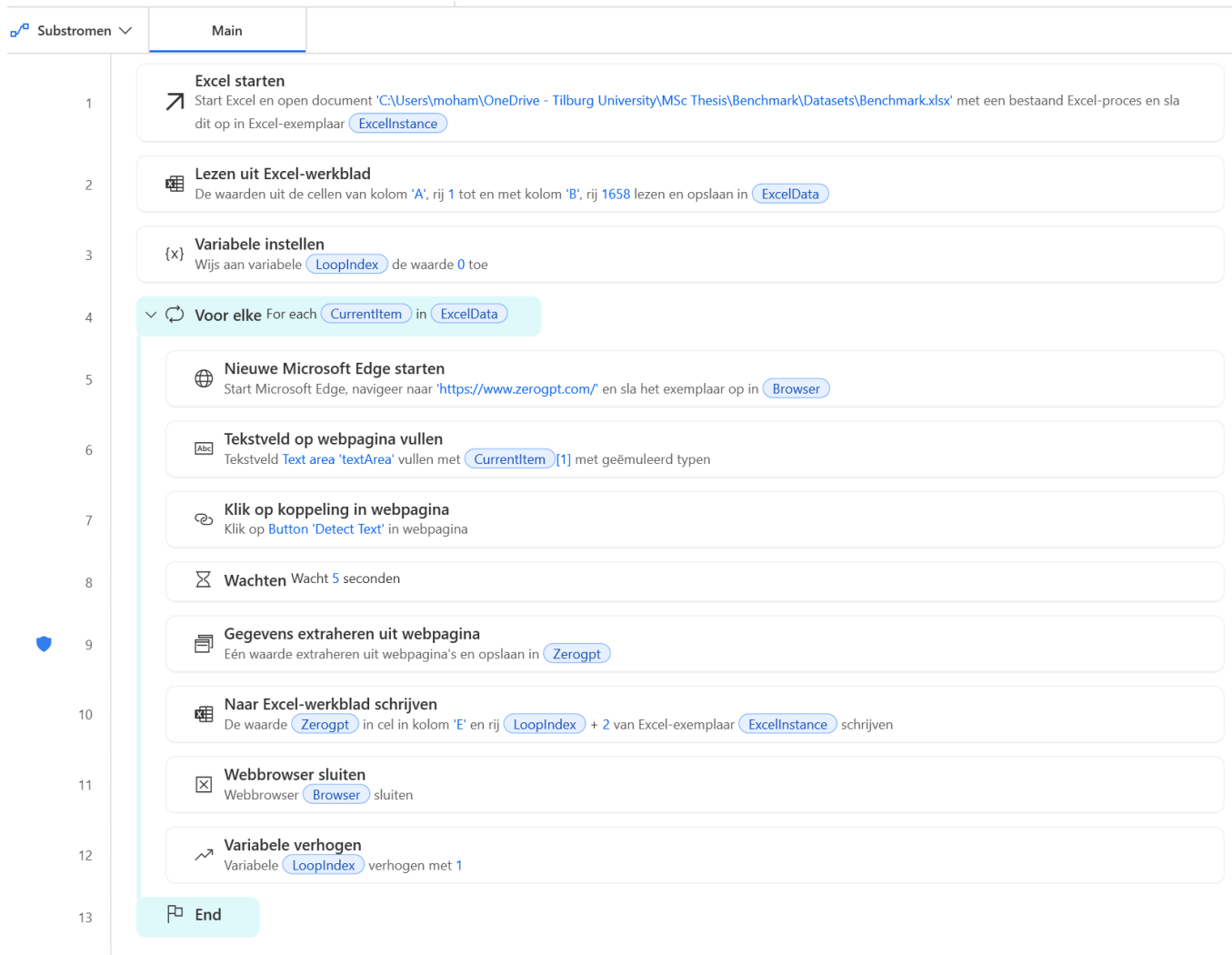
Total Visits

Dec 2024 - Feb 2025 Worldwide



B. Overview of the Power Automate Flow

Below is an example of a Power Automate flow that was created to benchmark the AI detection tools.



C. Interview Questions

Draft Interview Questions for Researchers on AI-Generated Text Detection on X (Twitter)

Introduction:

Thank you for taking the time to participate in this interview. My research focuses on detecting AI-generated text on X (formerly Twitter) and social media platforms.

I am particularly interested in the effectiveness of existing detection models, their limitations, and potential improvements. Your insights will be invaluable in understanding current methodologies and shaping future developments in this field.

General Background:

- What do you think about AI Detection tools in general?

Considerations

- How reliable and effective do you consider current AI detection tools in identifying AI-generated content? (*opinion*)

Improvements

- What key improvements are necessary to enhance the accuracy of these models?
- How do you foresee AI-generated text detection evolving in the next 5-10 years?

Closing Question:

- Is there anything else you believe is important to consider when researching AI-generated text detection on social media?

Ending:

Thank you for sharing your insights and expertise. Your perspective is invaluable in furthering research on AI-generated text detection. If you have any additional thoughts, recommendations, or relevant resources, please feel free to share them.

D. Python Code

This appendix presents the python code that was used for analyzing the result of this research.

PERFORMANCE METRICS OF AI DETECTION TOOLS

```
import pandas as pd
import numpy as np

# Total tweets per dataset
tweet_counts = {
    "GPT-2": 200,
    "GPT-3": 200,
    "GPT-4o": 200,
    "GPT-4o with emojis": 200,
    "Human": 800
}

# Raw results in percentages
results = {
    "Dataset": ["GPT-2", "GPT-3", "GPT-4o", "GPT-4o with emojis", "Human"],
    "Grammarly (%)": [20, 52, 98, 29, 3],
    "GPTZero (%)": [35, 96, 97, 74, 1],
    "ZeroGPT (%)": [12, 86, 96, 91, 4]
}

# Create DataFrame
df = pd.DataFrame(results)

# Convert percentages to numbers based on tweet counts
for tool in ["Grammarly (%)", "GPTZero (%)", "ZeroGPT (%)"]:
    df[tool.replace("(", "(count)")] = df.apply(lambda row: (row[tool] / 100) *
        tweet_counts[row["Dataset"]], axis=1)

print(df)

# Calculate TP, FP, TN, FN per tool
metrics = {}
```

```

tools = ["Grammarly", "GPTZero", "ZeroGPT"]

for tool in tools:
    ai_detected = df.loc[df["Dataset"] != "Human", f"{tool} (count)"].sum()
    human_detected = df.loc[df["Dataset"] == "Human", f"{tool} (count)"].values[0]

    TP = ai_detected
    FN = 800 - ai_detected # total AI tweets = 200 * 4 = 800
    FP = human_detected
    TN = 800 - human_detected # total Human tweets = 800

    metrics[tool] = {"TP": TP, "FP": FP, "TN": TN, "FN": FN}

# Display results
for tool, metric_values in metrics.items():
    print(f"{tool} results:")
    print(metric_values)

# Results per tool
results = {
    'Grammarly': {'TP': 398, 'FP': 24, 'TN': 776, 'FN': 402},
    'GPTZero': {'TP': 604, 'FP': 8, 'TN': 792, 'FN': 196},
    'ZeroGPT': {'TP': 570, 'FP': 32, 'TN': 768, 'FN': 230}
}

# Define functions to calculate metrics
def calculate_metrics(TP, FP, TN, FN):
    accuracy = (TP + TN) / (TP + FP + TN + FN)
    precision = TP / (TP + FP) if (TP + FP) != 0 else 0
    recall = TP / (TP + FN) if (TP + FN) != 0 else 0
    false_positive_rate = FP / (FP + TN) if (FP + TN) != 0 else 0
    false_negative_rate = FN / (FN + TP) if (FN + TP) != 0 else 0
    return accuracy, precision, recall, false_positive_rate, false_negative_rate

# Calculate metrics for each tool
metrics_summary = []

```



```
for tool, vals in results.items():
    accuracy, precision, recall, fpr, fnr = calculate_metrics(
        vals['TP'], vals['FP'], vals['TN'], vals['FN']
    )
    metrics_summary.append({
        'Tool': tool,
        'Accuracy': round(accuracy, 3),
        'Precision': round(precision, 3),
        'Recall (Sensitivity)': round(recall, 3),
        'False Positive Rate': round(fpr, 3),
        'False Negative Rate': round(fnr, 3)
    })
```

```
# Display results in a DataFrame
metrics_df = pd.DataFrame(metrics_summary)
print(metrics_df)
```

```
#### Visusualization of the results ###
import matplotlib.pyplot as plt
import numpy as np
```

```
# Define the metrics and scores for each tool
metrics = ['Accuracy', 'Precision', 'Recall']
grammarly = [0.73375, 0.9431, 0.4975]
gptzero = [0.8725, 0.9869, 0.7550]
zerogpt = [0.83625, 0.9468, 0.7125]
```

```
# Set position for each metric group
x = np.arange(len(metrics))
width = 0.25 # Width of the bars
```

```
# Create the plot
fig, ax = plt.subplots(figsize=(9, 6))
```

```
# Plot bars for each tool
ax.bar(x - width, grammarly, width, label='Grammarly', color='green')
ax.bar(x, gptzero, width, label='GPTZero', color='blue')
ax.bar(x + width, zerogpt, width, label='ZeroGPT', color='gray')

# Add labels and formatting
ax.set_xlabel('Performance Metrics')
ax.set_ylabel('Scores')
ax.set_title('Performance of Detection Tools by Metric')
ax.set_xticks(x)
ax.set_xticklabels(metrics)
ax.set_ylim(0, 1)
ax.legend()

# Add gridlines for better readability
ax.grid(axis='y', linestyle='--')

# Improve layout
plt.tight_layout()
plt.show()
```

Performance Metrics of Datasets

```
import pandas as pd

# Total tweets per dataset
tweet_counts = {
    "GPT-2": 200,
    "GPT-3": 200,
    "GPT-4o": 200,
    "GPT-4o with emojis": 200,
    "Human": 800
}

# Raw results in percentages
results = {
    "Dataset": ["GPT-2", "GPT-3", "GPT-4o", "GPT-4o with emojis", "Human"],
    "Grammarly (%)": [20, 52, 98, 29, 3],
    "GPTZero (%)": [35, 96, 97, 74, 1],
    "ZeroGPT (%)": [12, 86, 96, 91, 4]
}

# Create DataFrame
df = pd.DataFrame(results)

# Convert percentages to numeric counts
for tool in ["Grammarly (%)", "GPTZero (%)", "ZeroGPT (%)"]:
    df[tool.replace("%", "(count)")] = df.apply(lambda row: (row[tool] / 100) *
        tweet_counts[row["Dataset"]], axis=1)

# Display intermediate results
print("Converted Counts per Dataset:")
print(df[["Dataset", "Grammarly (count)", "GPTZero (count)", "ZeroGPT (count)"]])

# Define function to calculate metrics per dataset
def calculate_metrics_dataset(predicted_ai, actual_ai, total_tweets):
```

```

TP = predicted_ai if actual_ai else 0
FP = predicted_ai if not actual_ai else 0
FN = total_tweets - predicted_ai if actual_ai else 0
TN = total_tweets - predicted_ai if not actual_ai else 0

```

```

accuracy = (TP + TN) / total_tweets
precision = TP / (TP + FP) if (TP + FP) != 0 else 0
recall = TP / (TP + FN) if (TP + FN) != 0 else 0
fpr = FP / (FP + TN) if (FP + TN) != 0 else 0
fnr = FN / (FN + TP) if (FN + TP) != 0 else 0

```

```

return accuracy, precision, recall, fpr, fnr

```

```

# Calculate metrics per dataset, per tool
tools = ["Grammarly (count)", "GPTZero (count)", "ZeroGPT (count)"]
datasets = df["Dataset"]

# Human-generated dataset is "Human"
actual_ai_datasets = ["GPT-2", "GPT-3", "GPT-4o", "GPT-4o with emojis"]

```

```

metrics_result = []

```

```

for dataset in datasets:
    total = tweet_counts[dataset]
    actual_ai = dataset in actual_ai_datasets
    for tool in tools:
        pred_ai = df.loc[df["Dataset"] == dataset, tool].values[0]
        accuracy, precision, recall, fpr, fnr = calculate_metrics_dataset(pred_ai, actual_ai, total)
        metrics_result.append({
            "Dataset": dataset,
            "Tool": tool.replace(" (count)", ""),
            "Accuracy": accuracy,
            "Precision": precision,
            "Recall": recall,
            "False Positive Rate": fpr,

```

```

        "False Negative Rate": fnr
    })

# Convert results to DataFrame for clarity
df_metrics = pd.DataFrame(metrics_result)

# Display final results
print("\nPerformance Metrics per Dataset and Tool:")
print(df_metrics)

import matplotlib.pyplot as plt
import numpy as np

# Datasets and tools
datasets = ['GPT-2', 'GPT-3', 'GPT-4o', 'GPT-4o Emojis', 'Human']
tools = ['Grammarly', 'GPTZero', 'ZeroGPT']

# Metrics data
accuracy = np.array([
    [0.10, 0.26, 0.49, 0.145, 0.97], # Grammarly
    [0.175, 0.48, 0.485, 0.37, 0.99], # GPTZero
    [0.06, 0.43, 0.48, 0.455, 0.96] # ZeroGPT
])

precision = np.array([
    [1, 1, 1, 1, 0], # Grammarly
    [1, 1, 1, 1, 0], # GPTZero
    [1, 1, 1, 1, 0] # ZeroGPT
])

recall = np.array([
    [0.10, 0.26, 0.49, 0.145, 0], # Grammarly
    [0.175, 0.48, 0.485, 0.37, 0], # GPTZero
    [0.06, 0.43, 0.48, 0.455, 0] # ZeroGPT
])

```

```
# Function to create bar plots
def plot_metric(metric, metric_name):
    x = np.arange(len(datasets))
    width = 0.25

    fig, ax = plt.subplots(figsize=(10, 6))

    bars1 = ax.bar(x - width, metric[0], width, label='Grammarly', color='green')
    bars2 = ax.bar(x, metric[1], width, label='GPTZero', color = 'blue')
    bars3 = ax.bar(x + width, metric[2], width, label='ZeroGPT', color='gray')

    ax.set_xlabel('Datasets')
    ax.set_ylabel(metric_name)
    ax.set_title(f'{metric_name} of Detection Tools Across Datasets')
    ax.set_xticks(x)
    ax.set_xticklabels(datasets)
    ax.legend()
    ax.grid(axis='y', linestyle='--')

    plt.ylim(0, 1)
    plt.show()

# Plot each metric
plot_metric(accuracy, 'Accuracy')
plot_metric(precision, 'Precision')
plot_metric(recall, 'Recall')
```

T-TEST

```
import pandas as pd
from scipy.stats import ttest_ind
import numpy as np

# Create an empty DataFrame
df = pd.read_excel('raw data.xlsx', sheet_name='raw data', engine='openpyxl')

# Now perform independent t-tests between tools
results = []

# Compare Grammarly vs GPTZero
t_stat_gptzero, p_val_gptzero = ttest_ind(df['Grammarly Result'], df['GPTZero Result'],
equal_var=False)
results.append(['Grammarly vs GPTZero', t_stat_gptzero, p_val_gptzero])

# Compare Grammarly vs ZeroGPT
t_stat_zerogpt, p_val_zerogpt = ttest_ind(df['Grammarly Result'], df['ZeroGPT Result'],
equal_var=False)
results.append(['Grammarly vs ZeroGPT', t_stat_zerogpt, p_val_zerogpt])

# Compare GPTZero vs ZeroGPT
t_stat_gptzero_zerogpt, p_val_gptzero_zerogpt = ttest_ind(df['GPTZero Result'], df['ZeroGPT
Result'], equal_var=False)
results.append(['GPTZero vs ZeroGPT', t_stat_gptzero_zerogpt, p_val_gptzero_zerogpt])

# Make it a DataFrame for easier export
ttest_results = pd.DataFrame(results, columns=['Comparison', 't-statistic', 'p-value'])

# Save to CSV
ttest_results.to_csv('t_test_results.csv', index=False)

# Print results
print(ttest_results)
```


E. Summary of the interview transcripts

Interview with Dr. Giuseppe Cascavilla

Q1: What do you think about AI Detection tools in general?

AI detectors, especially those relying on traditional machine learning trained on separate human- and AI-generated datasets, are fundamentally unreliable. They don't generalize well and are prone to false positives and negatives.

Q2: How reliable and effective are current AI detection tools?

Current tools are not reliable. They rely heavily on surface-level patterns, which newer AI models are learning to mask. The tools are also often abandoned like a recent internal ChatGPT project due to limited long-term effectiveness.

Q3: What improvements are necessary?

Detectors would need to identify more abstract cues, such as uncommon word use or suspiciously generic phrasing. Even then, success is limited because generative models continuously improve and adapt.

Q4: How do you see AI detection evolving?

I do not foresee effective AI detectors becoming mainstream. The AI models are already too advanced and improving faster than detection systems.

Interview with Prof. Grzegorz Chrupala

Q1: Is AI detection on social platforms like X feasible?

It is theoretically possible to train detection models on platforms like Twitter, but it would require enormous effort, computational resources, and come with significant privacy challenges.

Q2: How can we detect AI-written text?

There are superficial signs like perfect grammar, marketing-like tone, or lack of spelling mistakes that may suggest AI authorship. But these are not definitive.

AI models have the style of marketing writing too.

Q3: What's the fundamental problem?

We are in an arms race: as detectors improve, so do generators. Static training data simply can't keep up.

Interview with Prof. Harm Brouwer

Q1: How widespread is AI-generated content on social media?

AI tools are widely used—students, professionals, and bots alike generate content with them. It’s nearly impossible to know who wrote what.

Q2: Are detection tools accurate?

Current detection tools are not trustworthy. They lack consistency and often fail to pick up nuances in AI-written text.

Q3: What makes AI-generated content difficult to detect?

AI-generated content can seem “hollow”—it sounds articulate but lacks substance. ChatGPT, for example, can explain theories but often invents incorrect citations or names.

Q4: What are the broader implications?

Writing is inherently a social act. If machines write for people, we lose touch with the intent and context. Plus, with AI now training on its own output, we are entering a feedback loop that may distort the internet’s content baseline.

Q5: Do you see effective detection tools coming soon?

Not likely. The gap between generative and discriminative models keeps widening, and that makes detection an increasingly elusive goal.

Interview with Prof. M. Alimardini

Q1: Can we rely on machine learning-based AI detectors?

Not really. These systems are easy to fool and often rely on outdated datasets. Once trained, their accuracy degrades as new models emerge.

Q2: What’s needed to improve detection?

Detection should go beyond just model output probability. It should look at metadata, authorship consistency, and even cross-platform behaviors.

Q3: Will social media platforms adopt AI detection?

Not in the short term. There are too many ethical and technical hurdles, especially with respect to user privacy and platform liability.

Interview with Anonymous Researcher

Q1: Can we realistically detect AI-written content?

It is highly unlikely we’ll ever have perfect detection tools. AI-generated text is becoming increasingly indistinguishable from human writing, particularly in short-form content like tweets.

Q2: What do you see as the main challenge?

Every improvement in detection is matched by an improvement in generation. And unlike earlier models, modern AIs produce text that mimics human error, randomness, and style almost perfectly.

Q3: Final thoughts?

Even if we develop an accurate detector today, it may become obsolete tomorrow. The nature of this field makes it difficult to create lasting solutions.